

Thermoelectric Properties of PEDOT:PSS

Zeng Fan and Jianyong Ouyang*

Nowadays, organic thermoelectric (TE) materials have attracted considerable attention due to their unique merits, e.g., light-weight, high mechanical flexibility, nontoxicity, easy availability, and intrinsically low thermal conductivity. Among the organic/polymer TE materials reported so far, poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is extensively investigated because it is water-processable, thermally stable, and can be highly conductive. Over the past few years, the TE properties of the PEDOT-based TE materials are continuously improved. With rational design, some PEDOT:PSS-based materials have achieved high ZT values comparable to the conventional inorganic TE materials like bismuth telluride at room temperature. This paper reviews the recent breakthroughs for PEDOT:PSS-based TE polymers and composites. The strategies for achieving high-performance PEDOT:PSS-based TE materials and the corresponding underlying mechanism are specifically discussed. The TE devices fabricated by the PEDOT:PSS-based TE materials are also presented, in terms of their fabrication/assembly technique, device configuration and device performance. With all the exciting progress made in the PEDOT:PSS-based TE materials, the further development and practical applications of the high-efficient organic TE materials as flexible TE module devices and wearable electronics can be greatly anticipated.

where S is the Seebeck coefficient, σ is the electrical conductivity, T is the absolute temperature, and κ is the thermal conductivity.^[4] Nowadays, the wide applications of TE materials in practice are still largely restricted by their relatively low energy conversion efficiency. It has been reported that, ZT values of at least 3.0 need to be achieved by a TE cooler for the substitution of the traditional compressor-based refrigerators, whereas ZT values above 2.0 are highly expected for effective power generation with an efficiency over 20%. As such, the development of TE materials with high ZT values becomes the primary and urgent academic and industrial challenge.

In the recent years, various inorganic semiconductors like bismuth (Bi)-tellurium (Te)-antimony (Sb)-selenium (Se) and nanostructures, e.g., superlattices, quantum dots, and nanoinclusions have achieved high ZT values above unity at room temperature.^[2,4–8] Nevertheless, their problems including the expensive-ness of the raw materials, complicate materials structure, toxicity of the scarce

1. Introduction

Thermoelectric (TE) materials, which enable the direct conversion between heat energy and electricity, have lots of potential for energy demands of the future. TE devices can be either used as power generators through the Seebeck effect, or be utilized as solid-state coolers through the Peltier effect.^[1–3] Due to their advantages, e.g., no moving parts, no working noise, long operating lifetime and easy scalability in size and power, TE materials have been receiving great attention. The energy conversion efficiency of a TE device is commonly defined by the dimensionless figure of merit (ZT) of the TE material, $ZT = S^2\sigma T/\kappa$,

elements and difficulty in manufacturing unfortunately restrict the sustainable development of these inorganic TE materials in the future.^[9,10] On the other hand, with the rapid development of miniaturized, portable, wearable, or implantable electronics, the desire for flexible and reliable power supply systems become even stronger than ever. To power these devices, the power generators should be conformal with deformation while maintaining their power supply capabilities. In this regard, the inorganic TE materials with intrinsic brittleness fail to meet the requirements arising from the highly flexible electronics. Therefore, great effort has been made in the development of organic TE materials as alternatives to inorganic TE materials.

The last two decades have witnessed amazing progress made on TE polymers and composites. In sharp contrast to the inorganic counterparts, organic TE materials, which refer to conducting polymers with a conjugated backbone structure, commonly possess unique advantages, including good mechanical flexibility, light weight, low cost, material abundance, low or no toxicity, etc.^[11–14] Most importantly, thermal conductivities of the organic materials can be at least one order lower than those of most inorganic semiconductors and semimetals. Given the significant impact of the thermal conductivity term on the overall ZT , the intrinsic low thermal conductivity of polymeric materials is of vital importance for the optimization of their TE properties. Polyacetylene (PA) has been reported

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to have an ultrahigh power factor of $\approx 1350 \text{ W m}^{-1} \text{ K}^{-2}$, but it has no practical significance due to its poor processability and poor stability. Currently, conducting polymers, e.g., poly(3,4-ethylenedioxythiophene):poly(styrene sulfonate) (PEDOT:PSS), polythiophene derivatives, polyaniline (PANI), and polypyrrole (PPy), are representative p-type organic semiconductors, which have been receiving the most extensive investigations as TE polymers. Among these conducting polymers, PEDOT:PSS is particularly attractive due to its merits including easy doping tunability, high transparency, high mechanical flexibility, excellent thermal stability, and facile solution-processability. In comparison with other conducting polymers like PANi and PPy, the superior TE property of PEDOT:PSS may be attributed to the good planarity and high conjugated degree in its backbones.

As shown in **Figure 1**, PEDOT:PSS consists of positively charged conjugated PEDOT chains and negatively charged PSS chains, for which the PEDOT chains are attached to the PSS chains through the Columbic interaction. PEDOT is a derivative of polythiophene, which can be prepared by the oxidative polymerization of its monomer ethylenedioxythiophene (EDOT) via either an electrochemical or a chemical polymerization approach. PSS is a polymer surfactant, which not only serves as the counterions for charge-balancing but also allow the dispersion of PEDOT in the aqueous environment. In order to stabilize the PEDOT:PSS in water, excessive poly(4-styrenesulfonic acid) (PSSH) is added to the dispersion. The ratio of PEDOT to PSS in the aqueous dispersion is usually 1:2.5 by weight.

PEDOT:PSS has been regarded as the most successful conducting polymer in terms of applications. There have been several grades of PEDOT:PSS aqueous dispersion available on the market.^[15] Owing to the good stability of the dispersion, continuous and uniform PEDOT:PSS films can be readily formed on substrates by various facile solution-processing techniques, such as coating, casting, spraying, and printing. However, an as-prepared PEDOT:PSS film from its aqueous dispersion usually has very low TE properties. The Seebeck coefficient and electrical conductivity are only $15\text{--}18 \mu\text{V K}^{-1}$ ^[16–19] and $0.2\text{--}1 \text{ S cm}^{-1}$,^[15,20,21] respectively. Correspondingly, its power factor ($\text{PF} = S^2\sigma$), which is widely adopted to evaluate the TE property of conducting polymers, is only $\approx 0.01 \mu\text{W m}^{-1} \text{ K}^{-2}$. Therefore, the as-prepared PEDOT:PSS materials cannot be directly utilized for TE applications. However, many methods have been developed to improve the TE properties of PEDOT:PSS in terms of both the Seebeck coefficient and electrical conductivity.

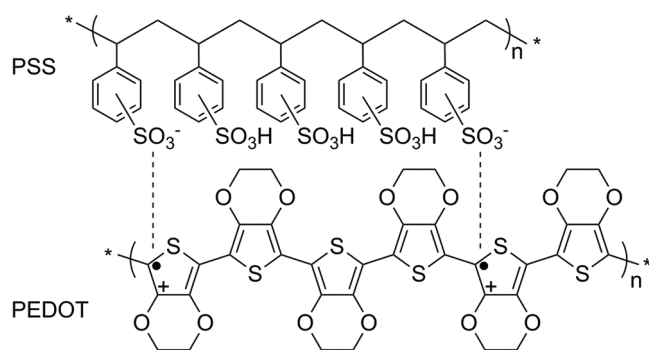
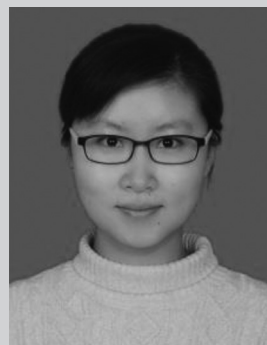


Figure 1. Chemical structure of PEDOT:PSS.



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In this review, we focus on the recent progress made on the organic TE materials based on PEDOT:PSS and its composites. We start from the preparation of the PEDOT-based polymers and composites, and then discuss the strategies and underlying mechanism for their TE property enhancement. Several representative TE devices fabricated by the PEDOT-based TE materials are also summarized in terms of their fabrication/assembly technique, configuration, and device performance. Finally, a perspective and outlook for the further development of PEDOT:PSS-based TE materials is discussed.

2. TE Properties of PEDOT:PSS and PEDOT:PSS-Based Composites

2.1. PEDOT:PSS

2.1.1. Secondary Doping

Secondary doping has been the most common approach for enhancing the electrical conductivity of PEDOT:PSS since

2000s or even earlier time.^[22] As the PF and ZT values are proportional to electrical conductivity, a significant increase in the electrical conductivity of PEDOT:PSS can directly lead to the great enhancement of its TE property. Secondary doping can be conducted by either adding a secondary dopant to the PEDOT:PSS aqueous suspension, or post-treating an as-prepared PEDOT:PSS film with the secondary dopant.^[23,24] The role of the secondary dopant is to improve the PEDOT:PSS morphology and thus improve the charge transport.^[25–28] Secondary doping generally hardly affects the doping level of PEDOT:PSS.^[22,27,28]

Organic polar solvents are widely investigated as the secondary dopants for the conductivity enhancement of PEDOT:PSS.^[27,29,30] For example, Luo et al. investigated the effect of two organic solvents, dimethyl sulfoxide (DMSO) and ethylene glycol (EG), on the TE properties of the PEDOT:PSS films.^[16] By adding DMSO or EG into the PEDOT:PSS aqueous dispersion, the electrical conductivities of the spin-coated PEDOT:PSS films were enhanced from 0.2 to $\approx 900 \text{ S cm}^{-1}$, while their Seebeck coefficients did not change saliently. As a result, PFs of ≈ 30 and $\approx 25 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$ were achieved for the DMSO- and EG-doped PEDOT:PSS films, respectively.

Gong and co-workers added binary dopants consisting of DMSO and poly(ethylene oxide) (PEO) to PEDOT:PSS and prepared the PEDOT:PSS films by spin coating.^[31] For the first time, they observed the simultaneous enhancements in both the Seebeck coefficient and electrical conductivity. The PEDOT:PSS films doped with 5% DMSO and 3% PEO can possess a high Seebeck coefficient of $40 \text{ } \mu\text{V K}^{-1}$, and electrical conductivity of 1300 S cm^{-1} , and an optimal PF of $157.4 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$. They ascribed the TE property enhancement to the conversion of polarons to bipolarons which can be tuned by PEO of the binary secondary dopants.^[12,32–34]

Although adding solvent additives to the PEDOT:PSS aqueous dispersion has been widely adopted as an effective method to improve the conductivity of the PEDOT:PSS films, the enhancements in their electrical conductivity still turn out to be limited due to the presence of large amount of excessive PSS.^[35] In order to remove these excessive insulating PSS components, post-treatment methods of PEDOT:PSS films were developed as an alternative approach.^[36] It has been revealed that polar organic solvents with a dielectric constant of 35–40 are particularly effective in enhancing the conductivity of PEDOT:PSS films.^[37] Xu and co-workers studied the effects of several polar solvents including *N*-methyl-2-pyrrolidone, EG, DMSO, and *N,N*-dimethylformamide (DMF) on the TE properties of PEDOT:PSS films.^[38] The EG-treated PEDOT:PSS films could exhibit the maximum PF of $14.9 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$ with a Seebeck coefficient of $17 \text{ } \mu\text{V K}^{-1}$ and an electrical conductivity of 520 S cm^{-1} . The enhancement by the post-treatment with organic polar solvents is usually ascribed to the PSS segregation and PEDOT conformational change.

The organic polar solvents can weaken the Coulombic interaction between PEDOT and PSS via a screening effect, thus facilitating phase segregation of PSS from PEDOT on the microscopic scale.^[39,40] Accompanied with the phase segregation, the PEDOT chains also transit from a coil to a linear or expanded coil conformation.^[29] Wei and co-workers studied the morphological change of the PEDOT grains before and after EG treatment by

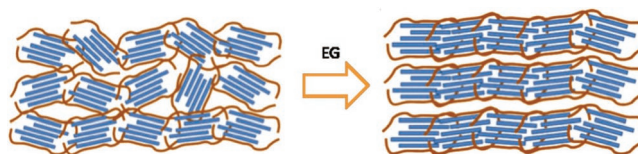


Figure 2. Schematic diagram for the morphological change in the PEDOT:PSS films by secondary doping. Reproduced with permission.^[41] Copyright 2013, Wiley-VCH.

the grazing-incidence wide angle X-ray diffraction and grazing-incidence small-angle X-ray scattering.^[41] The crystallinity of PEDOT and the ordering of the PEDOT nanocrystals along the through-plane are improved after the secondary doping process (Figure 2). The carrier mobility of the PEDOT:PSS films remarkably increases from 0.045 to $1.7 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$.

Apart from neat solvents, cosolvents consisting of two or more kinds of organic solvents have also been used for the post-treatment.^[41,42] Deng and co-workers investigated the post-treatment of PEDOT:PSS films with the cosolvent of EG and DMSO.^[18] However, their highest PF was only $37.05 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$ for the PEDOT:PSS treated with neat DMSO. Notably, although the less polar organic solvents, e.g., ethanol, isopropyl alcohol (IPA), acetonitrile (ACN), DMF, and acetone are not able to significantly improve the conductivity of PEDOT:PSS, their cosolvents with water appear to be quite effective.^[43] After the post-treatment with a cosolvent, e.g., DMF/water, γ -butyrolactone/water, acetone/water, ACN/water, methanol/water, ethanol/water, and IPA/water, the electrical conductivities of the PEDOT:PSS films were remarkably enhanced by 3–4 orders of magnitude.^[43,44] However, their Seebeck coefficients only slightly altered in a small range of 20.6 – $25.1 \text{ } \mu\text{V K}^{-1}$. A maximum PF of $52.5 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$ was obtained for the PEDOT:PSS films treated with the cosolvent of 80% DMF-20% water. For these cosolvent-treated PEDOT:PSS films, the great enhancement in their TE property was mainly due to the phase segregation of their PEDOT and PSS chains. Because the hydrophobic PEDOT and hydrophilic PSS chains were preferentially solvated with the organic solvent and water in the cosolvent, respectively, the Columbic attraction between the positively charged PEDOT and negatively charged PSS could be effectively screened.^[43] As shown in Figure 3, for the PEDOT:PSS films treated with cosolvents, the conductivity enhancement well correlated with the dielectric constant of the organic solvent used.^[44]

Compared with the treatments with neat organic solvents, post-treatments with solutions as the secondary dopants can give rise to even more significant TE property enhancement for PEDOT:PSS.^[37,45–47] For instance, organic solutions of inorganic salts like zinc chloride (ZnCl_2) and copper (II) chloride (CuCl_2) were able to enhance the TE properties of the PEDOT:PSS films with PFs exceeding $50 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$.^[48] After a treatment with the DMF solution of ZnCl_2 , the Seebeck coefficient and electrical conductivity were simultaneously improved. Under the optimal conditions, the treated PEDOT:PSS films exhibited a Seebeck coefficient of $26.1 \text{ } \mu\text{V K}^{-1}$ and electrical conductivity of 1400 S cm^{-1} , resulting in a power factor of $98.2 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$ at room temperature. The mechanism for such a great enhancement in the TE properties can be attributed to the synergetic effects of

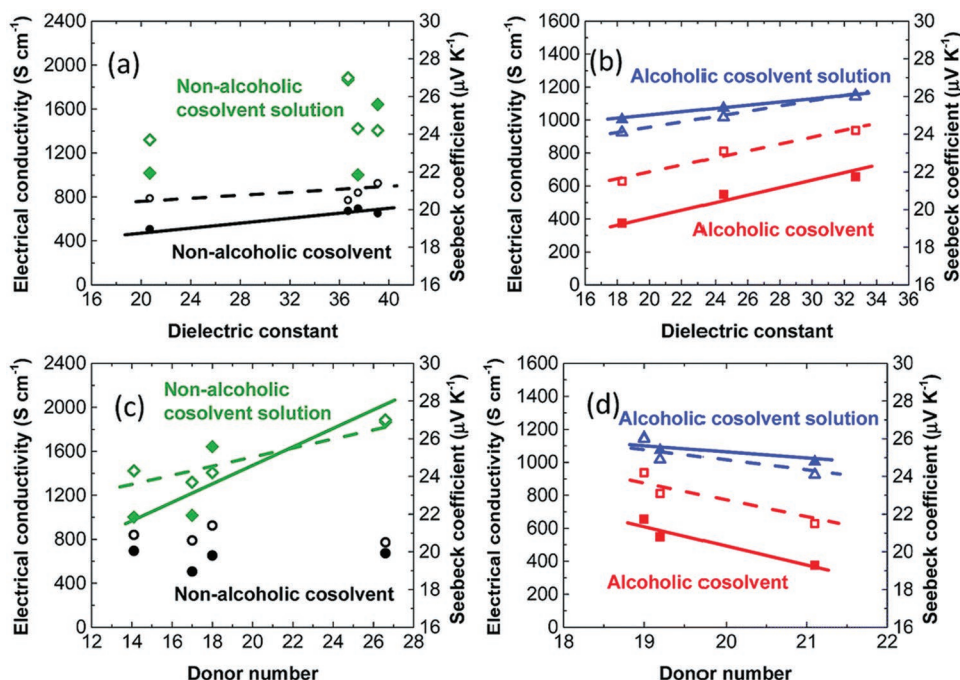


Figure 3. Effects of a,b) the dielectric constant and c,d) the donor number of the organic solvent on the TE properties of the cosolvent and cosolvent solution-treated PEDOT:PSS films. All panels reproduced with permission.^[44] Copyright 2018. Royal Society of Chemistry.

inorganic salts and organic solvents that effectively screen the Coulombic interaction between PEDOT and PSS. The post-treatment with a solution not only facilitates effective depletion and phase segregation of the PSS chains from PEDOT:PSS but also induces conformational change of the PEDOT chains from a benzoid to a quinoid structure (as shown in Figure 4).

Solutions consisting of an organic salt dissolved in cosolvents were also used to treat the PEDOT:PSS films.^[44] As the treatment with cosolvent solutions possessed both synergetic effects arising from the organic solvent/water and the salt/cosolvent, its effects on the TE property enhancement became more pronounced over the treatment with neat cosolvents

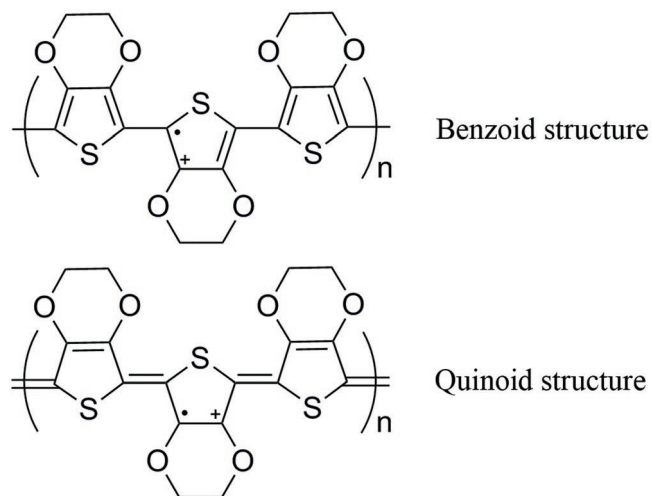


Figure 4. Benzoid and quinoid structures of PEDOT. Reproduced with permission.^[29] Copyright 2004, Elsevier.

or organic solutions with neat solvent. For instance, the PEDOT:PSS films treated with a cosolvent solution of 1 M methylammonium iodide (MAI) in 80% DMF-20% water successfully reached a PF of $144\ \mu W\ m^{-1}\ K^{-2}$ with a Seebeck coefficient of $28\ \mu V\ K^{-1}$ and electrical conductivity of $1831\ S\ cm^{-1}$, respectively. The PF was among the highest value reported for the PEDOT:PSS films after a one-step post-treatment. In addition, the TE property enhancement of PEDOT:PSS after the treatment with cosolvent solutions was found to be highly dependent on the physical property of the organic solvents. As shown in Figure 3, for the cosolvent solution consisting of a nonalcoholic organic solvent, the donor number of the solvent was closely related to its capability for the TE property enhancement of PEDOT:PSS. For the cosolvent solution consisting of an alcoholic organic solvent, the dielectric constant of the alcohol dominated the TE property of PEDOT:PSS after treatment.^[44]

In addition to polar organic solvents or solutions, various acids, including organic carboxylic acids and inorganic acids, etc. have also been demonstrated to effectively enhance the TE properties, especially the electrical conductivity of PEDOT:PSS.^[20,49–54] Chu and co-workers treated the vacuum-filtered PEDOT:PSS bulky papers with various secondary dopants, e.g., EG, polyethylene glycol, methanol, and formic acid (FA).^[55] They found that the PEDOT:PSS bulky paper treated with FA exhibited the best TE performance. They observed a Seebeck coefficient of $20.6\ \mu V\ K^{-1}$ and an electrical conductivity of $1900\ S\ cm^{-1}$, and a high PF of $80.6\ \mu W\ m^{-1}\ K^{-2}$. Using the Harman method, the in-plane and through-plane ZT values of the FA-treated PEDOT:PSS bulky paper were measured to be 0.12 and 0.32, respectively. They are among the highest values for PEDOT:PSS.

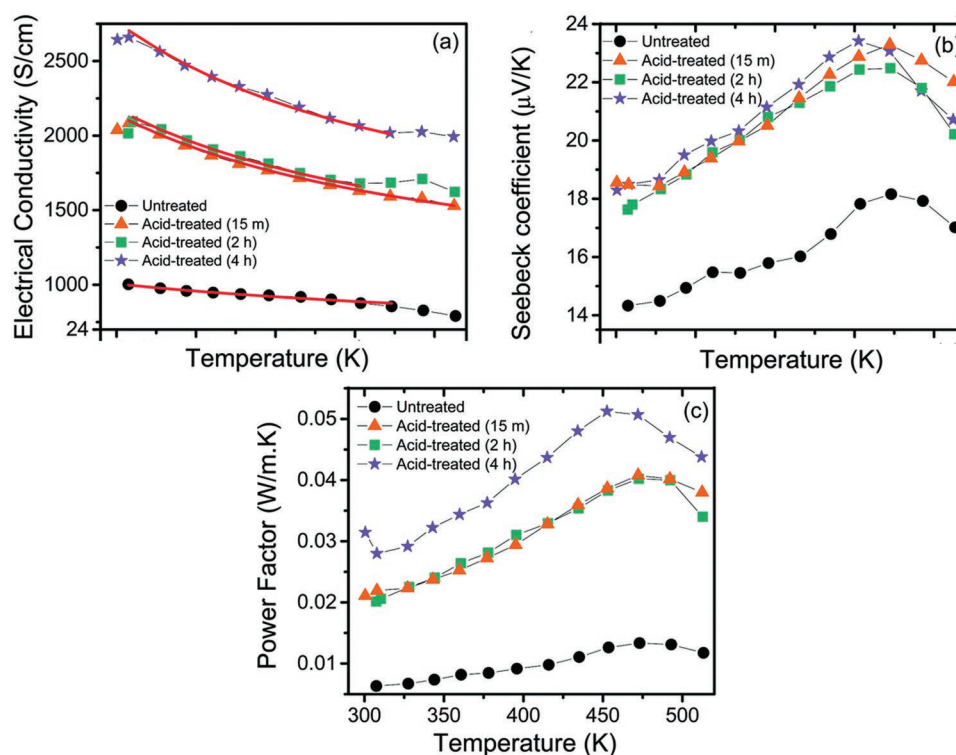


Figure 5. TE properties of H₂SO₄-treated PEDOT:PSS films as a function of temperature. a) Electrical conductivity, b) Seebeck coefficient, and c) power factor. All panels reproduced with permission.^[56] Copyright 2016, Royal Society of Chemistry.

Alshareef and co-workers treated the methanol-added PEDOT:PSS films with sulfuric acid (H₂SO₄) and explored the temperature-dependent TE performance of the untreated and treated PEDOT:PSS films.^[56] They found that the acid treatment not only significantly enhanced the electrical conductivities of the PEDOT:PSS films but also induced a noticeable increase in the Seebeck coefficient. As a result, the power factor was improved by a factor of 5 compared with that of the pristine films. The enhancements are attributed to the loss of PSS from the PEDOT:PSS films and the morphology change of the PEDOT grains after the H₂SO₄ treatment. As shown in **Figure 5**, both the Seebeck coefficient and electrical conductivity of the treated films increased with the elevating temperature up to 460 K. At 460 K, the maximum power factor multiplied by the absolute temperature ($S^2\sigma T$) reached $0.052 \text{ W m}^{-1} \text{ K}^{-1}$.

2.1.2. Chemical Dedoping

In terms of the fundamental physics, the Seebeck coefficient of a TE material depends on the density of states (DOS).^[57] Through either a physical or a chemical approach, tuning doping levels that can impact both the band structure and the DOS of the TE materials has shown to be a versatile approach for tailoring the Seebeck coefficient.^[58] A pristine PEDOT:PSS film has a doping level of 33%, corresponding to a carrier concentration of 10^{21} – 10^{22} cm^{-3} . Dedoping has been widely proven to be an effective approach for the enhancement of Seebeck coefficient. Lowering the carrier concentration in PEDOT:PSS can enhance its Seebeck coefficient at a sacrifice of its electrical

conductivity. In spite of the trade-off between the Seebeck coefficient and electrical conductivity, there was still an optimal doping level that can remarkably improve the PF and ZT. For instance, Crispin and co-workers reported the dedoping of PEDOT:PSS by varying the gate voltage in an organic electrochemical transistor configuration.^[59] As shown in **Figure 6**, the Seebeck coefficient increased while the electrical conductivity decreased for the PEDOT:PSS films with the increasing gate voltage. As the doping level was lowered from 33% to 14.5%, an optimal power factor of $23.5 \mu\text{W m}^{-1} \text{ K}^{-2}$ was obtained.

Treating PEDOT:PSS films with chemical reducing agents are also widely used for the TE property enhancement. PEDOT:PSS films dedoped with common reducing agents, e.g., hydrazine, ammonium formate (AF), sodium sulfite (Na₂SO₃), tetrakis(dimethylamino)ethylene (TDAE), sodium borohydride (NaBH₄), hydroiodic acid (HI), and 1-ethyl-3-methylimidazolium tetrafluoroborate (EMIM-BF₄) have been reported.^[19,60–64] Carella et al. treated the μm -thick PEDOT:PSS films with several common reducing agents (as listed in **Table 1**).^[62] There is a correlation between the Seebeck coefficient enhancement and the redox potentials for the aqueous solutions of reducing agents. After a treatment with the strongest reducing agent NaBH₄, the PEDOT:PSS films exhibited a high Seebeck coefficient up to $141 \mu\text{V K}^{-1}$. However, the electrical conductivity was significantly decreased to only 0.1 S cm^{-1} .

Luo et al. studied the TE properties of the PEDOT:PSS thin films which were treated with DMSO solution of EMIM-BF₄.^[61] As the EMIM-BF₄ concentration increased from 0 to 100 vol%, the Seebeck coefficient of the films increased from 15.7 to $23 \mu\text{V K}^{-1}$, whereas the conductivity decreased

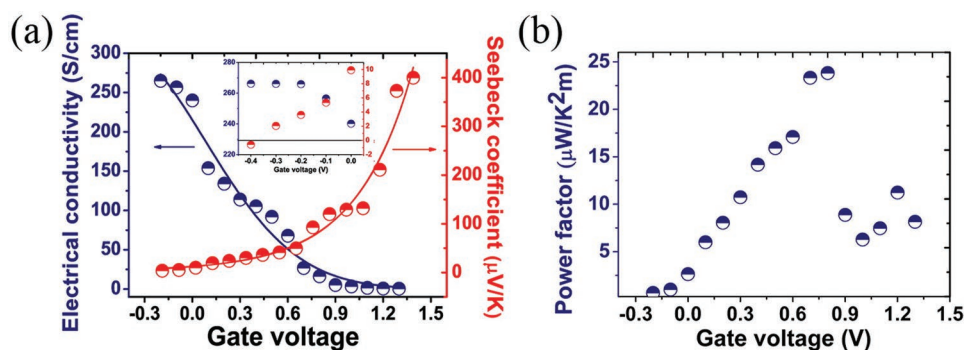


Figure 6. TE properties of the PEDOT:PSS films as a function of the gate voltage in the organic electrochemical transistor configuration. a) Electrical conductivity (blue symbols) and Seebeck coefficient (red symbols) and b) power factor. All panels reproduced with permission.^[59] Copyright 2011, American Chemical Society.

from 900 to 700 S cm⁻¹. When the concentration of EMIM-BF₄ was controlled to be 50 vol% in DMSO, an optimal PF of 38.46 μW m⁻¹ K⁻² was achieved. For the films treated in the presence of EMIM-BF₄, the formation of short and circular PEDOT grains was observed (Figure 7a). The transition of PEDOT chains from bipolaron to polaron states was characterized by the UV-vis absorption spectra. The neutral, polaron and bipolaron states of the PEDOT chains show absorption intensities at the wavelengths of 600 nm, 900 nm, and the near infrared region, respectively.^[65,66] As shown in Figure 7b, the increased intensity at 900 nm for the EMIM-BF₄-treated PEDOT:PSS films indicated the partial dedoping of PEDOT from bipolaron to polaron state.

Apart from the redox reactions, the TE properties of PEDOTs can also be optimized by tuning their acidity through an acidobasic treatment.^[67–69] In an acid environment, the proton ions dissociated from a photonic acid can usually be added to the backbones of the conducting polymers, thereby giving rise to the effect as protonic acid doping.^[11,70] Such protonic doping effect highly depends on the pH level. Increasing the pH can lead to the removal of these protons, which is called deprotonation, similar to the effect of dedoping. Stepien et al. directly modified to the PEDOT:PSS aqueous solution with a strong base, potassium hydroxide (KOH).^[71] The concentration of KOH was varied from 0 × 10⁻³ to 160 × 10⁻³ M, corresponding to the pH level change from 1.8 to 13. By preparing the PEDOT films with these modified PEDOT:PSS dispersions,

they found that the Seebeck coefficient of the as-prepared films dramatically increased from 15 to 90 μV K⁻¹ while the electrical conductivity decreased from 110 S cm⁻¹ to 0.02 S m⁻¹ with the increasing PH.

Later, Wei et al. reported a novel solid-state photoinduced reaction to dedope PEDOT:PSS.^[72] Figure 8 illustrated the photoinduced dedoping reaction. A PEDOT:PSS film was first prepared via spin-coating and then immersed in a commercially available water-soluble photobase generator 2-(9-oxoxanthene-2-yl) propionic acid 1,5,7-triazabicyclo [4.4.0]dec-5-ene salt (PBG). When the PBG is irradiated at 365 nm, PBG can transit to 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD), which is a strong base having a pK_a of 26. As a result, the positively charged PEDOT can be dedoped to the neutral state by accepting the electrons from TBD. The carrier density in the PEDOT:PSS films was finely tuned by variation of the UV-irradiation time. At the optimal carrier concentration of ≈ 5 × 10²⁰ cm⁻³, the film gained a maximum PF of ≈ 42 μW m⁻¹ K⁻².

2.1.3. Sequential Treatments

As mentioned above, the secondary doping methods can significantly increase the electrical conductivity of PEDOT:PSS while only slightly affect the Seebeck coefficient. Dedoping can remarkably enhance the Seebeck coefficient but severely decrease the electrical conductivity. In order to combine the enhanced electrical conductivity arising from secondary dopants and the improved Seebeck coefficient facilitated by dedoping, sequential treatment processes have been developed. For example, Kim and co-workers prepared the PEDOT:PSS films from the 5% DMSO-added PEDOT:PSS aqueous dispersion, and then further treated the films by overcoating a mixture of DMSO and hydrazine onto the film surface.^[60] As shown in Figure 9a, by controlling the concentration of hydrazine in DMSO, they observed an increase in the Seebeck coefficient from 30 to 142 μV K⁻¹ and a decrease in electrical conductivity from 726 to 2 S cm⁻¹, leading to an optimal PF of 112 μW m⁻¹ K⁻². In such a two-step process, the addition of 5% DMSO first changed the polymer conformation from a coiled to a linear or expanded-coil structure, while the post-treatment with the DMSO solution of hydrazine subsequently dedoped the PEDOT to a lower oxidation level (Figure 9b).

Table 1. TE properties of the PEDOT:PSS films treated with aqueous solutions of several typical reducing agents. Adapted with permission.^[62] Copyright 2014, Royal Society of Chemistry.

	Reducing agent	Redox potential E ⁰ [V/SHE]	Seebeck coefficient [μV K ⁻¹]	Electrical conductivity [S cm ⁻¹]
1	None	N/A	18	714.2
2	Na ₂ S ₂ O ₃	-0.57	59	234.7
3	TDAE	-0.71	43	3.7
4	Na ₂ SO ₃	-0.93	70	139.7
5	Hydrazine	-1.16	92	0.1
6	NaBH ₄	-1.24	104	0.1

Note: After treatment, the PEDOT:PSS films were sequentially rinsed in EG and water baths.

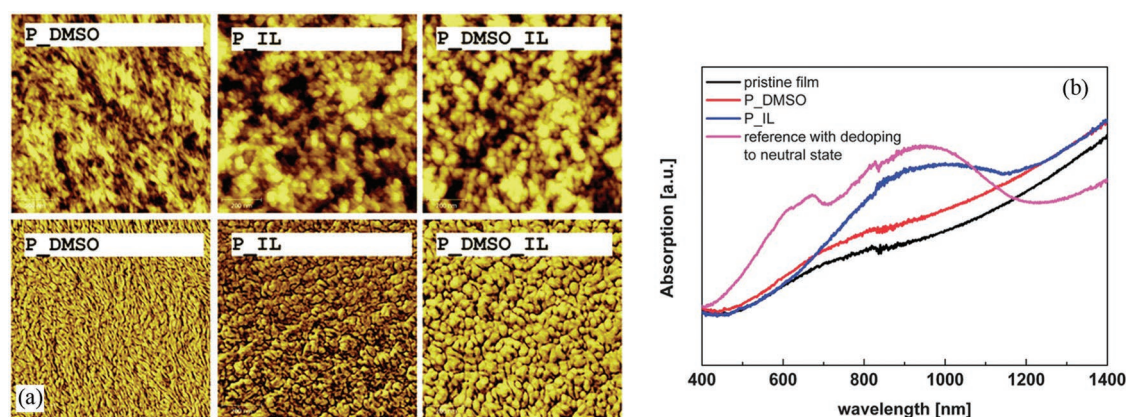


Figure 7. a) AFM images and (b) UV-vis absorption spectra of the PEDOT:PSS films treated with DMSO, ionic liquid (IL) and the DMSO solution of IL. IL donates for the EMIM-BF₄ which is an ionic liquid. All panels reproduced with permission.^[61] Copyright 2013, Royal Society of Chemistry.

Later, Kim et al. modified the sequential treatment approach by changing the secondary dopant from DMSO to the DMSO solution of *p*-toluenesulfonic acid monohydrate (TSA).^[73] As the TSA enabled additional photonic doping to the PEDOT backbone thiophenes in addition to the conformation change of polymer chains, the electrical conductivity of the PEDOT:PSS films after the secondary doping reached 1218 S cm⁻¹. In comparison, the electrical conductivity of the DMSO-doped PEDOT:PSS film was 970 S cm⁻¹. When the subsequent DMSO/hydrazine was then used to treat the DMSO/TSA-added PEDOT:PSS films, the oxidation level of PEDOT was further tuned. When the carrier concentration decreases from 8×10^{22} to 4×10^{22} cm⁻³, the Seebeck coefficient increases from 24.1 to 50.4 μ V K⁻¹ while the electrical conductivity decreases from 1647 to 783 S cm⁻¹. At the optimal conditions, the PEDOT:PSS films can achieve a PF of 318.4 μ W m⁻¹ K⁻².

Recently, Sun and co-workers carried out a systematic investigation on a few representative sulfonic acids as the secondary dopant by adding into the PEDOT:PSS aqueous dispersion.^[74]

They found that the conductivity enhancement of the as-formed PEDOT:PSS films was highly affected by the steric effect and conjugated structure of these acid dopants. When doped with benzenesulfonic acid (BSA), the PEDOT:PSS films could achieve an electrical conductivity of up to 1996 S cm⁻¹. After the BSA-doped film was subsequently subjected to a treatment with DMSO/hydrazine solution, its doping level was well optimized. As the hydrazine concentration increased from 0 to 30 vol% in DMSO, the Seebeck coefficient increased from 18.1 to 39.1 μ V K⁻¹, accompanied with the decrease in electrical conductivity from 3049 to 812 S cm⁻¹. The maximum PF of 203.1 μ W m⁻¹ K⁻² was obtained when the electrical conductivity and Seebeck coefficient were 1119 S cm⁻¹ and 42.6 μ V K⁻¹, respectively.

The TE properties of PEDOT:PSS films can also be significantly enhanced through sequential post-treatments. Fan et al. treated the pristine PEDOT:PSS films with common acid, e.g., H₂SO₄ and common base, e.g., sodium hydroxide (NaOH) in sequence.^[75] Due to the fact that the acid treatment significantly enhanced the electrical conductivity of the films, while the base treatment finely tuned the oxidation level of PEDOT, the optimal PEDOT:PSS films delivered a maximum PF of 334 μ W m⁻¹ K⁻² with the Seebeck coefficient of 39.2 μ V K⁻¹ and electrical conductivity of 2170 S cm⁻¹ at room temperature. This PF remains to be the world-record value ever achieved for the PEDOT:PSS TE films.

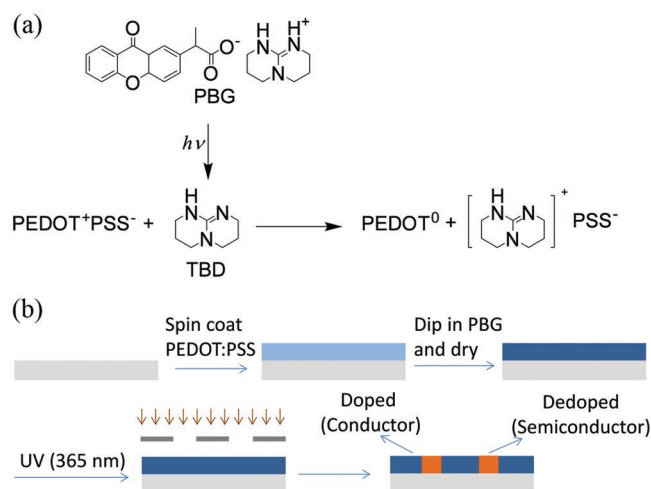


Figure 8. a) Photoinduced dedoping reaction and b) schematic diagram of the patterning process for the fabrication of the dedoped PEDOT:PSS/film via the photoinduced reaction. Reproduced with permission.^[72] Copyright 2016, American Chemical Society.

2.1.4. Morphology Control and Effect of Molecular Weight

For conducting polymers, the polymer chains can arrange either in an amorphous or a highly ordered fashion. The degree of the packing order in the polymer chains has also been found to greatly impact the TE properties of a conducting polymer.^[76–79] As shown in Figure 10a, the electrical conduction through PEDOT:PSS can involve the intrachain or interchain charge transport. As the crystallinity of PEDOT:PSS increases, the polymer chains become more aligned. These adjacent packed chains can result in the overlap of the π -electrons, thus facilitating the delocalization of the electronic wave function and highly promoting the charge transport across the crystalline domains.^[80,81] Figure 10b shows the electronic structure of

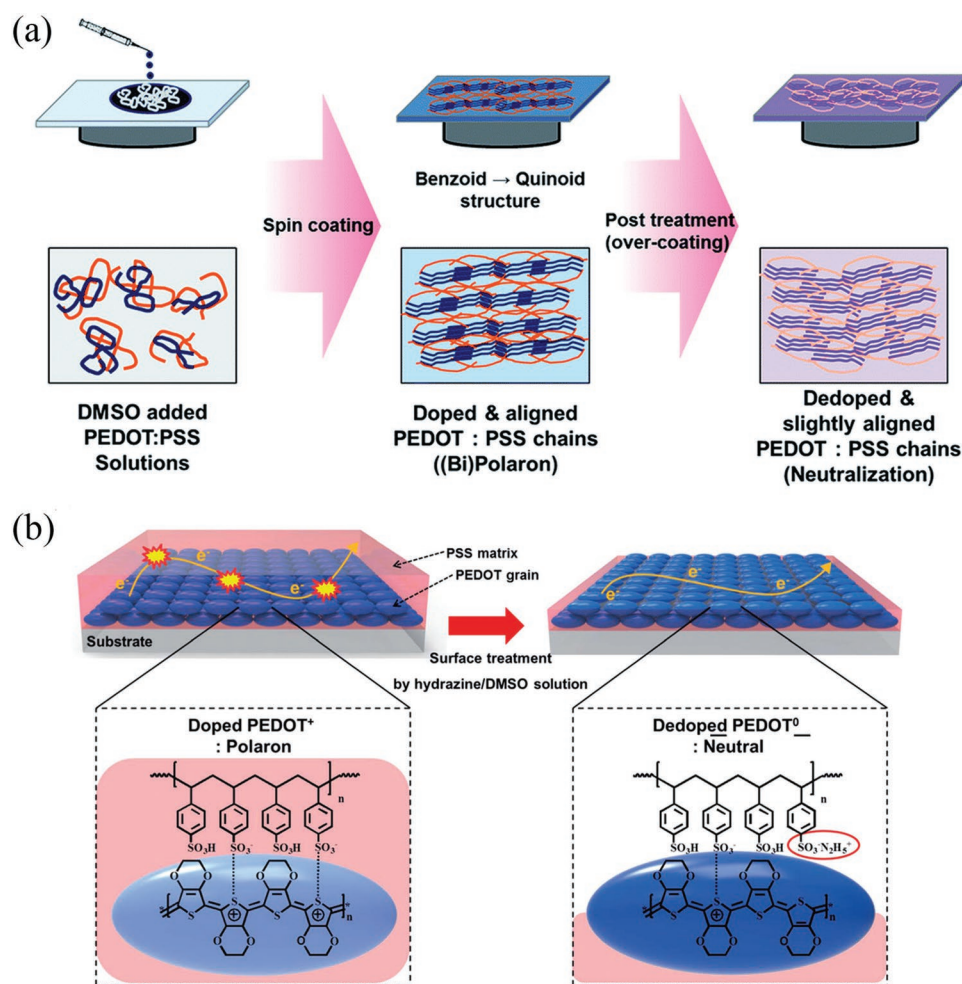


Figure 9. Schematic diagrams of a) a typical sequential treatment process of the PEDOT:PSS films. Reproduced with permission.^[60] Copyright 2014, Royal Society of Chemistry. b) The morphological change and doping state transition in the PEDOT:PSS films after sequential post-treatments. Reproduced with permission.^[73] Copyright 2014, Royal Society of Chemistry.

a typical conducting polymer like PEDOT.^[33] For the polymer in an amorphous phase, the Fermi level is supposed to lie among the localized states between the valence band and the bipolaron band. As this is a smooth region of the DOS, the slope of $\ln N(E)$ at the E_F is small, corresponding to a quite low Seebeck coefficient at low electrical conductivity. However, with the increasing chain packing order, an extended empty bipolaron band can be created and merge with the filled valence band. The slope of $\ln N(E)$ at E_F therefore becomes remarkably larger, which can in turn result in a high Seebeck coefficient at high electrical conductivity.^[12] However, the band structures of conducting polymers are largely inferred from their temperature-dependent transport and photoemission spectra.^[58] Much work still remains to be conducted to uncover their unique charge conduction physics.

Chen and co-workers investigated the TE performance of PEDOT in different nanostructures, for which the nanostructures were controlled by manipulating the polymerization rate during the chemical oxidation polymerization.^[77] They found that the Seebeck coefficient and electrical conductivity followed the same sequence as bulk PEDOT < globular PEDOT

nanoparticle < PEDOT nanorod < PEDOT nanotube < PEDOT nanofiber. The simultaneously enhanced Seebeck coefficient and electrical conductivity of these nanostructured PEDOTs are different from the trade-off between Seebeck coefficient and electrical conductivity for traditional inorganic semiconductors. As mentioned above, the expanded molecular conformation and ordered chain alignment can reduce the barrier for interchain charge hopping. For organic TE materials, carrier conduction in the highly disordered domains is assumed to be thermally assisted hopping, while carrier conduction in the highly ordered domains occur by band conduction.^[58] By means of tuning the ordered structure of PEDOT chain packings, the charge transport behavior accompanying with the polymer chain conformation and DOS at E_F is also modulated. In the case of a highly asymmetric DOS achieved at the E_F , the highly ordered PEDOTs could possibly show decoupled or even simultaneously enhanced Seebeck coefficient and electrical conductivity. However, the uncertainties in the DOS shape of conducting polymers still restrict a definitive demonstration of the charge transport mechanism. Other than the chain packing, factors including the polymer chain conformation, doping

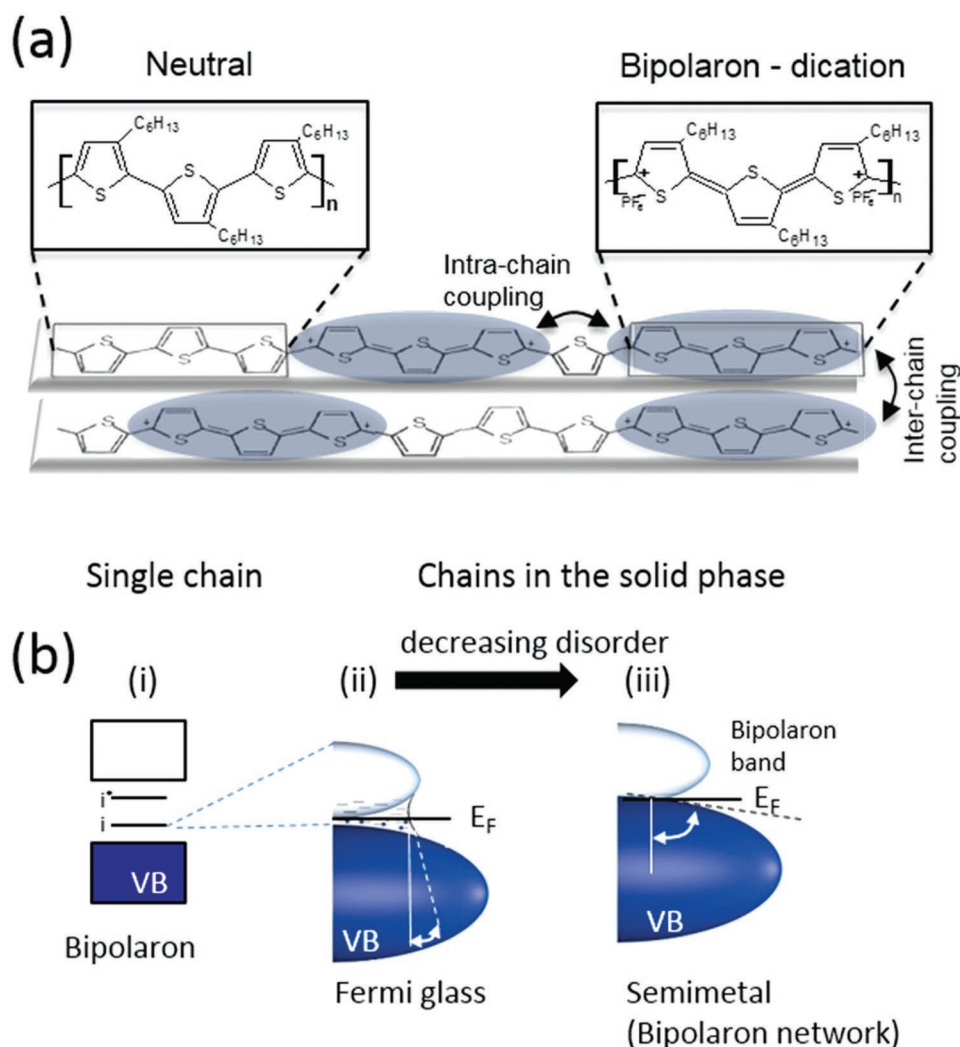


Figure 10. Schematic diagrams of a) the charge transport mechanism in the polythiophenes and b) the electronic structure of a typical conducting polymer. Reproduced with permission. Copyright 2016, Wiley-VCH. [80]

and oxidation levels, the polymer conjugation length and the interparticle contacts could also account for the enhanced TE properties.

Recently, Choi and co-workers reported that the addition of 4 wt% polyethylene glycol (PEG) into the PEDOT:PSS dispersion could improve the stretchability of the PEDOT:PSS films up to 50%. [82] They prepared PEDOT:PSS films added with 4% PEG directly on a highly stretchable elastomer, polydimethylsiloxane. The films were then subjected to a tensile strain of 30%. After stretching, a structural change of the PEDOT:PSS films from a dense network state to an isolated state with thick cracks was observed. Surprisingly, the morphological change lowers the electrical conductivity while enhance the Seebeck coefficient for the strained PEDOT:PSS. The PF of the PEDOT:PSS at the 30% stain was $47.3 \mu\text{W m}^{-1} \text{K}^{-2}$, which was evidently higher than that of the unstretched film ($35.1 \mu\text{W m}^{-1} \text{K}^{-2}$).

By comparing the TE properties of the PEDOT:PSS films prepared from two different grades of the commercial PEDOT:PSS aqueous dispersion (i.e., PH1000 and Clevios P), Fan et al. investigated the effect of PEDOT molecular weight on the TE

properties of PEDOT:PSS. [83] The Clevios PH1000 with higher PEDOT weight was found to constantly show higher Seebeck coefficient and electrical conductivity than the Clevios P. Although the post-treatments can remarkably enhance the PFs of both grades of the PEDOT:PSS film by a factor of 1.2–3.6, the highest PF of the Clevios PH1000 film after a treatment could be 30 times higher than that of the Clevios P counterpart. This result indicated the great importance of the PEDOT molecular weight on the TE property of PEDOT:PSS.

2.1.5. Surface Energy Filtering

In addition to the well-known electronic TE effect, significant ionic effect has also been revealed in PEDOT:PSS. As PSSH is a good ionic conductor, it is appropriate to take PEDOT:PSS as a mixed ionic-electronic conductor in nature. [84] The TE property of an ionic conductor or the ionic contribution in a mixed ionic-electronic conductor can be evaluated by the figure of merit ZT_i arising from the ionic conduction. It is defined as

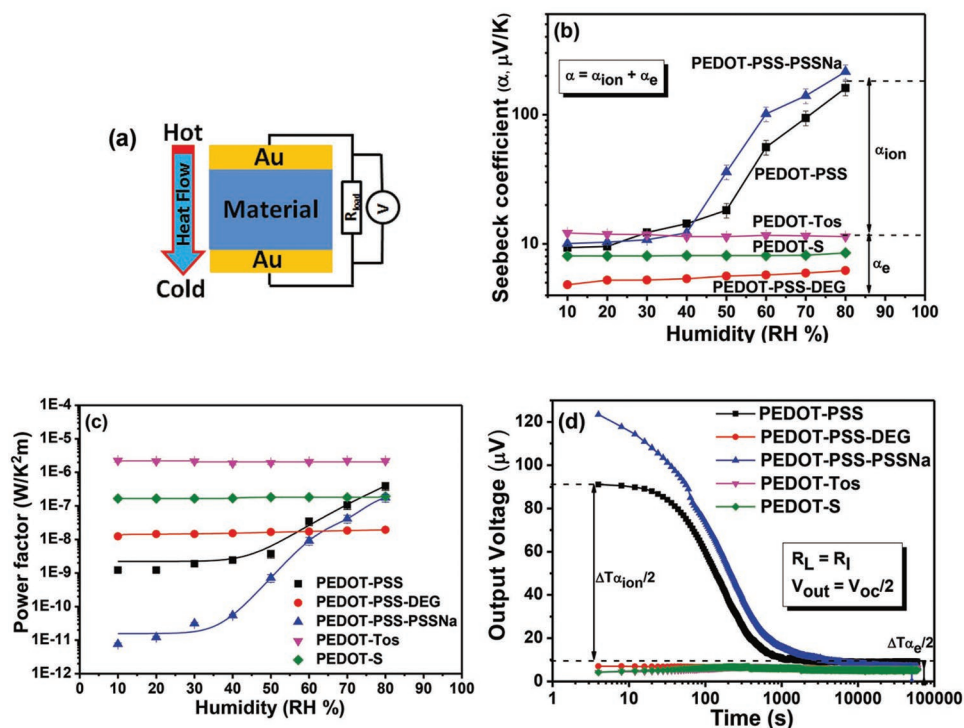


Figure 11. TE properties of several different types of polymeric conductors. a) Schematic diagram of the Seebeck coefficient measurement configuration. b) Seebeck coefficient and c) power factor of these polymeric conductors as a function of RH. d) Output voltage over the external load for these conductors versus time. The RH level is 80%. All panels reproduced with permission.^[85]

$ZT_i = S_i^2 \sigma_i T / \kappa_i$, where S_i , σ_i , and κ_i are the Seebeck coefficient, electrical conductivity, and thermal conductivity of the ionic portion, respectively.

Crispin and co-workers reported that the TE characteristics of the ionic and electronic charge carriers were significantly distinguishable under ΔT . When a polymeric conductor is connected to an external load, the output voltage generated by ions decays quickly after reaching the maximum value, while that by electrons is constant. Accordingly, for a mixed ionic-electronic conductor, the thermal generated voltage was a result of the sum of the ionic and electronic contributions. As an elevated relative humidity (RH) can promote ionic transport in solid polymeric conductors, the Seebeck coefficients of the ion-containing PEDOT:PSS like the pristine PEDOT:PSS and PEDOT:PSS-poly(sodium 4-styrenesulfonate) (PSSNa) were found to increase from ≈ 10 to over $100 \mu\text{V K}^{-1}$ at 80% RH (Figure 11). Their PFs were correspondingly enhanced by two and five orders of magnitude, respectively. At high RH levels, the large Seebeck coefficients are attributed to the thermodiffusion of heavy cations dissociated from the ionic conductors, which is called the Soret effect. However, the ion transport could be blocked at the interface with the metal electrodes, therefore not being able to pass across the external circuit. This specific thermodiffusion behavior of the cations eventually led to the remarkable voltage or current drop over the external load with a prolonged period of time.

Crispin and co-workers further conducted a comprehensive comparison on the TE characteristics of a pure electronic conductor like PEDOT:Tos (Tos = tosylate), a pure ionic conductor

like PSSNa and a mixed electronic-ionic conductor like the pristine PEDOT:PSS.^[86] Their time dependence behavior of the open circuit-voltage (V_{OC}) was proposed as an appropriate way to distinguish these three types of polymeric conductors, as shown in Figure 12a. At $\Delta T = 1 \text{ K}$, the V_{OC} of an electronic conductor (PEDOT:Tos) instantaneously followed the temperature rise and saturated at a constant value of several μV within only 30 s, whereas that of the ionic conductor increased much slower than the temperature rise but could reach a high voltage in the mV range after a period of $\approx 100 \text{ min}$. The mixed ionic-electronic conductor possessed a complex evolution of V_{OC} with time, which was different from either the pure electronic or the pure ionic one. The V_{OC} of the mixed conductor PEDOT:PSS first rapidly increased to the maximum value within 30 s with the increasing ΔT , and then gradually dropped to a much lower static value in the next few minutes. The V_{OC} at the peak value was strongly dependent on the RH levels (as shown in Figure 12b), indicating that the evolution of V_{OC} versus time should originate from a contribution of ion transport, as reported previously.

Figure 12c illustrated the complex feature of V_{OC} with time for the mixed conductor. Since the pristine PEDOT:PSS possesses a large amount of both holes and protons, significant thermodiffusion of both charge carriers can take place when a ΔT is applied. The holes are capable of showing a real-time response to the ΔT , whereas the protons build the concentration gradient at a much slower pace. As a result, the effectively instantaneous response is determined by the very different mobilities of holes and ions charges. In the first stage (region

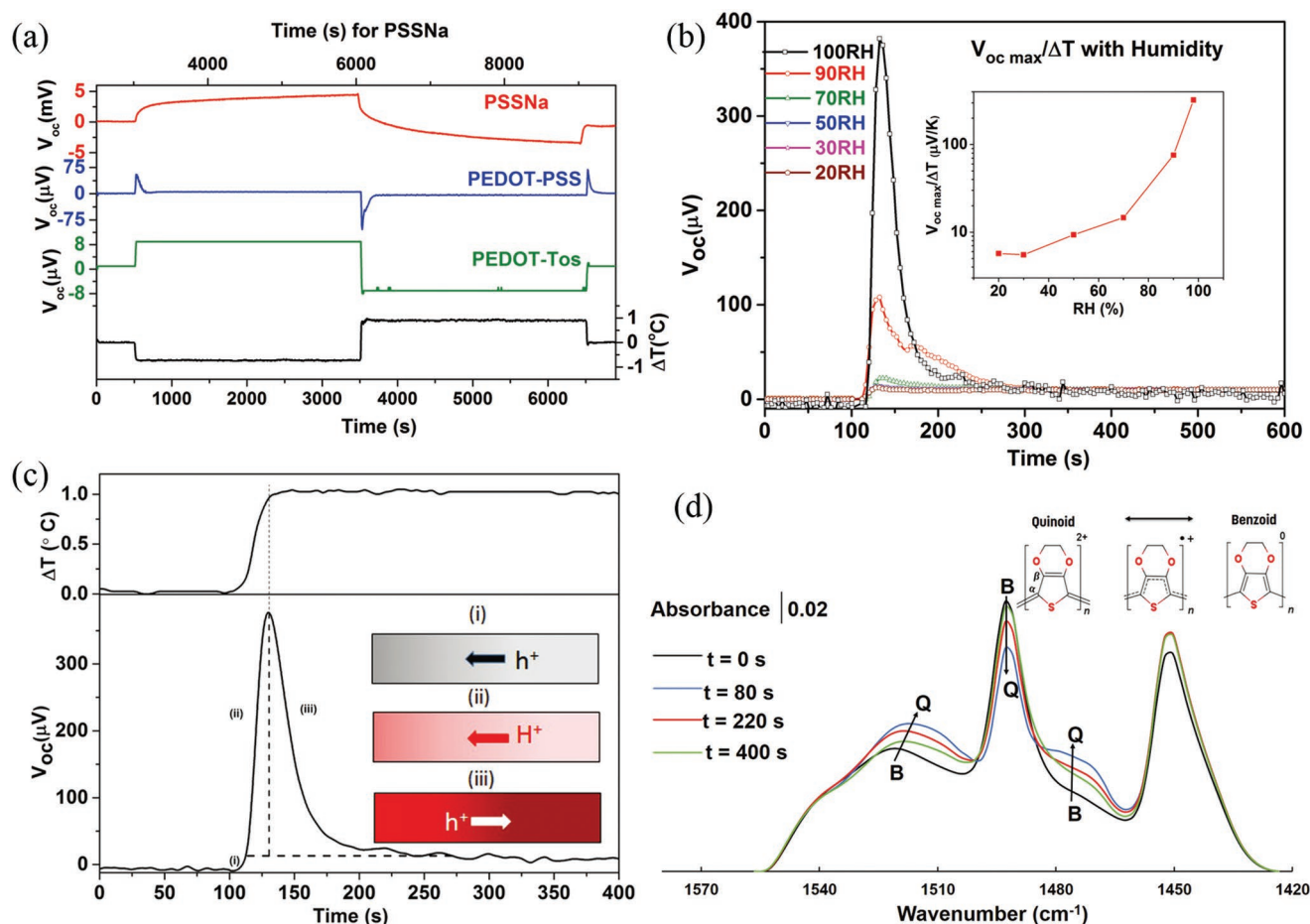


Figure 12. A comparison of the TE properties of three different polymeric conductors, i.e., PEDOT-Tos, PEDOT:PSS, and PSSNa. a) The evolution of their V_{OC} versus time at 80% RH with $\Delta T = 1$ K. b) Peak value of V_{OC} for the mixed ionic-electronic conductor PEDOT:PSS as a function of RH. c) The mechanism proposed for explaining the evolution of V_{OC} -time dependence for the mixed ionic-electronic conductor. d) FT-IR spectra showing the dynamic transfer process of holes and ions at ΔT . All panels reproduced with permission.^[86]

(i)), the fast increase of V_{OC} can be simply attributed to the thermodiffusion of holes arising from the Seebeck effect, while in the following stage (region (ii)), the ionic contribution just starts to be pronounced. Since the ionic Seebeck coefficient can be greater than the electronic Seebeck coefficient by several orders of magnitude, a remarkably high V_{OC} value is observed. Due to the as-built proton gradient, an internal electric field from the cold end toward the hot end is created. This internal electric field leads to an opposite hole drift current, which in turn limits the thermodiffusion of protons and thereby decreases the V_{OC} (region (iii)). The dynamic transfer process of the electronic and ionic carriers at ΔT was revealed by the in situ FT-IR spectra, as shown in Figure 12d. However, more work is needed to understand the charge transport of systems consisting of both electrons and ions in details.

Although the ionic or mixed ionic-electronic conductors are able to produce ultrahigh thermovoltages of several hundreds or thousands of $\mu V K^{-1}$ upon temperature gradients due to the Soret effect, their ionic conductivities are in general extremely low ($< 1 S cm^{-1}$).^[87–90] Consequently, the PFs of the ionic conductors are less than $2 \mu W m^{-1} K^{-2}$. Besides, the aforementioned ion-containing conductors usually exhibit decaying

output performance when connected to the external circuit, thus making their behavior like a capacitor rather than an electronic conductor.^[91–93]

Very recently, Fan et al. reported a novel approach to significantly enhance the TE performance of PEDOT:PSS by ion accumulation of an ionic liquid (IL) on the polymer surface.^[94] Heterostructures of PEDOT:PSS with ILs were constructed by directly coating an IL on the PEDOT:PSS film. The ion accumulation arising from the IL coating can significantly enhance the Seebeck coefficient of the PEDOT:PSS films by 1.2–2-fold, while it only slightly lowers the electrical conductivity (Figure 13a–c). As a result, the IL/PEDOT:PSS heterostructures exhibited a maximum power factor of $754 \mu W m^{-1} K^{-2}$ and ZT of 0.75. This ZT value is comparable to that of the conventional bismuth telluride (Bi_2Te_3) at room temperature.^[95] Importantly, the IL/PEDOT:PSS heterostructures exhibited significantly different V_{OC} evolution with time in comparison with the aforementioned mixed ionic-electronic conductors. As shown in Figure 13d, the V_{OC} of the IL/PEDOT:PSS heterostructures all instantaneously followed the ΔT increasing process, and no peak or drop appeared once the ΔT was stabilized. Interestingly, the IL/PEDOT:PSS heterostructure also have a constant

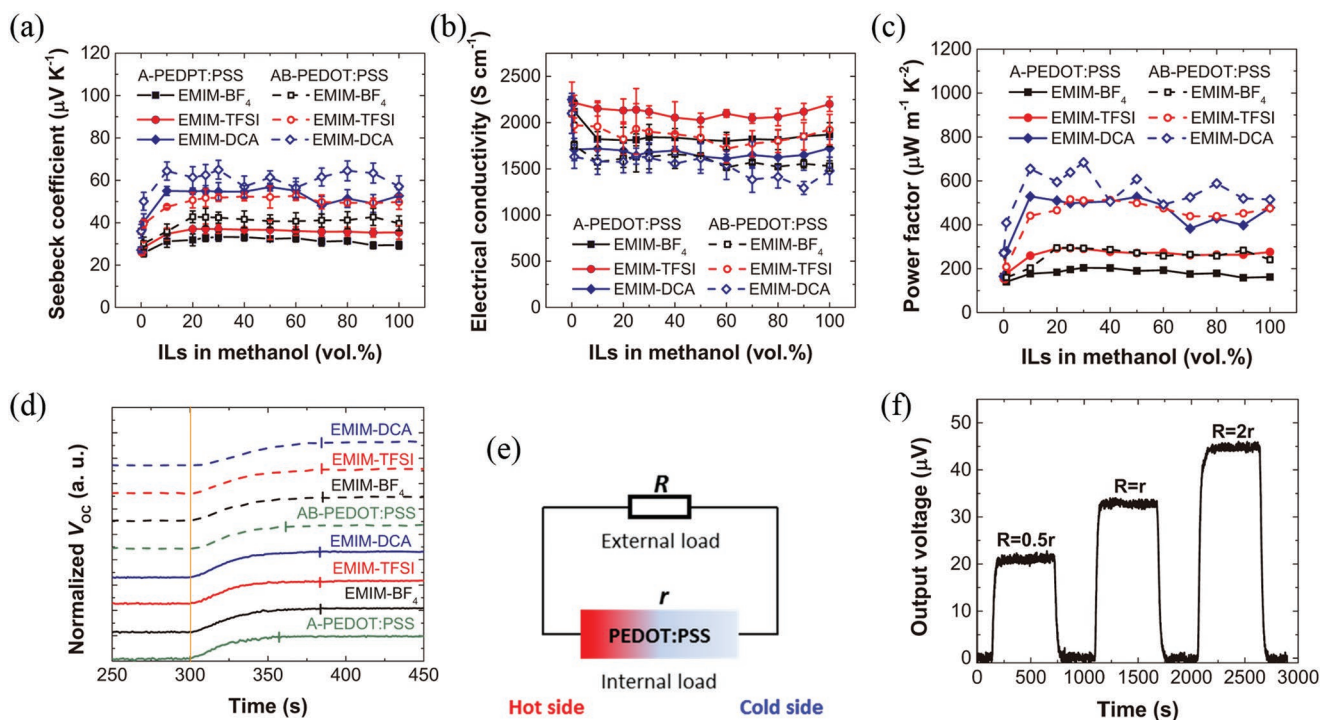


Figure 13. TE properties, time-dependent V_{OC} , and output performance of the IL/PEDOT:PSS heterostructure. a) Seebeck coefficient, b) electrical conductivity, and c) power factor of IL/PEDOT:PSS heterostructures as a function of the IL concentration in methanol. d) V_{OC} versus time of the IL/PEDOT:PSS heterostructures at $\Delta T = 1$ K. Solid and dashed lines represent the heterostructures with acid-treated PEDOT:PSS (A-PEDOT:PSS) and acid-base treated PEDOT:PSS (AB-PEDOT:PSS) films, respectively. EMIM-BF₄, 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMIM-TFSI) and 1-ethyl-3-methylimidazolium dicyanamide (EMIM-DCA) are ILs. e) Schematic diagram of the output voltage measured across an external resistor. f) Output voltage over the external resistor. All panels reproduced with permission.^[94] Copyright 2018, Elsevier.

voltage/current output over the external resistor, as shown in Figure 13e,f. The stable output performance suggested that the thermovoltage of the ionic conductor/conducting polymer heterostructures is produced by the holes in conducting polymers rather than the ions of the ionic conductors. The electronic behavior is substantially different from that of an ionic conductor of a mixed ionic-electronic conductor.

Later, the outstanding TE performance of such an ionic conductor/conducting polymer heterostructure was further verified on the heterostructures constructed by coating polyelectrolyte on the surface of the PEDOT:PSS film.^[96] A layer of polyelectrolyte, PSSH or PSSNa was coated on the highly conductive PEDOT:PSS film. Due to the fact that PSSH and PSSNa are solid electrolytes at ambient conditions, the effect of PSSH or PSSNa coating was sensitive to the RH. As shown in Figure 14a–c, as the RH level increased, the Seebeck coefficient of the PSSNa/PEDOT:PSS increased while its electrical conductivity slightly decreased. As a result, the PF also increased with increasing RH. The heterostructure consisting of PSSNa and PEDOT:PSS sequentially treated with an acid and base achieved the maximum PF of $401 \mu\text{W m}^{-1} \text{K}^{-2}$ at 100% RH.

One possible mechanism for the enhancement in the Seebeck coefficient by polyelectrolytes was illustrated in Figure 14d. In such a heterostructure, the cations dissociated from the electrolyte and holes of PEDOT:PSS are separately present in the upper and lower layer, respectively. Both of the cations

and holes will undergo thermodiffusion at ΔT . At the equilibrium state, the accumulated cations in the electrolyte layer can generate potential barriers along the surface of PEDOT:PSS, which prevent the transport of the low-energy holes and only allow the high-energy holes to diffuse toward the cold end. In terms of the definition of Seebeck coefficient, for which $S = (E_F - E_T)/T$, where T is the absolute temperature and E_T and E_F are the mean energy of the transporting charges and the Fermi energy, respectively,^[97–99] the Seebeck coefficient enhancement by polyelectrolytes can be ascribed to the increased mean hole energy in PEDOT:PSS. Future perspectives, e.g., the exploration of more appropriate models for explaining the remarkable TE property enhancement of such heterostructures, the characterization of the PEDOT conformation change induced by ionic coating, and the investigation of the interaction at the PEDOT/ionic conductor interfaces are still highly needed.

The TE properties of some typical PEDOT:PSS reported in literature are summarized in Table 2. The secondary doping methods can mainly enhance the electrical conductivity, and the dedoping can mainly enhance the Seebeck coefficient. The energy filtering can enhance the Seebeck coefficient while it only slightly affects the electrical conductivity.

Notably, the characterization techniques for the organic thin film TE materials like PEDOT:PSS are still immature compared with the well-established measurement standards for the inorganic bulk counterparts. For electrical conductivity

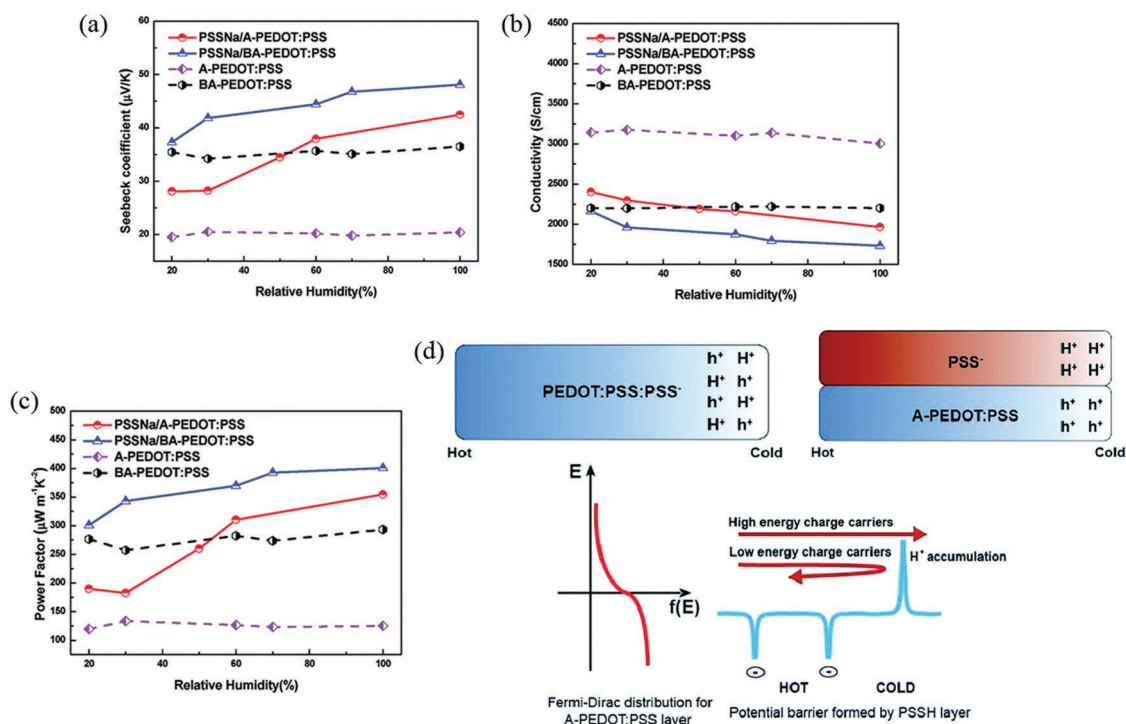


Figure 14. a–c) TE properties of the PEDOT:PSS films and the polyelectrolyte/PEDOT:PSS heterostructures as a function of RH. (a) Seebeck coefficient, (b) electrical conductivity, and (c) power factor. d) Schematic diagram of the energy filtering effect arising from the Soret effect of the ions in polyelectrolytes. A-PEDOT:PSS and BA-PEDOT:PSS represent the H_2SO_4 -treated and H_2SO_4 -NaOH treated PEDOT:PSS films, respectively. All panels reproduced with permission.^[96] Copyright 2018, Royal Society of Chemistry.

Table 2. A summary of TE properties of some typical PEDOT:PSS at room temperature.

Treatment		S [$\mu\text{V K}^{-1}$]	σ [S cm^{-1}]	PF [$\mu\text{W m}^{-1} \text{K}^{-2}$]	κ [$\text{W m}^{-1} \text{K}^{-1}$]	ZT at r.t.	Ref.
Secondary doping	Addition of DMSO and PEO	38.4	1061	157.35	–	–	[31]
	80% DMF-20% water	22.3	1055	52	0.29	0.05	[44]
	80%DMF-20% water of MAI	28	1831	144	0.29	0.11	
	Post treatment with DMF solution of ZnCl_2	26.1	1400	98.2	–	–	[48]
	Post treatment with FA	20.6	1900	80.6	–	–	[55]
	H_2SO_4	≈ 14.5	1000	≈ 21	–	–	[56]
Dedoping	Electrochemical dedoping	≈ 100	≈ 20	23.5	0.17	0.041	[59]
	Treated with NaBH_4	104	0.1	0.1	–	–	[62]
	Chemical dedoping with AF	436.3	0.036	0.69	0.32	1.1×10^{-3}	[64]
	Treated with EMIM- BF_4 and DMSO	≈ 23	≈ 700	38.46	0.17	≈ 0.068	[61]
	Treated with DMSO then DMSO and hydrazine	≈ 41	677	115.48	≈ 0.17	0.2	[63]
	Treated with NaOH	≈ 15.8	≈ 800	≈ 19.6	–	–	[69]
	Photoinduced dedoping	≈ 27	590	42	–	–	[72]
Sequential treatments	Addition of DMSO then treated with HZ	67	578	112	0.366 ± 0.016	0.093	[60]
	Treated with TSA and then hydrazine/DMSO	≈ 50	≈ 1300	318.4	0.3	0.31	[73]
	Treated with H_2SO_4 and then NaOH	39.2	2170	334	–	–	[75]
	Addition of BSA and then hydrazine	42.6	1119	201.3	–	–	[74]
Heterostructures	IL/PEDOT:PSS	≈ 69	≈ 1580	754	0.3	0.75	[94]
	PSSH/PEDOT:PSS	43.5	2120	401	–	–	[96]
	PSSNa/PEDOT:PSS	48.1	1732	401	–	–	

measurements, the four-probe techniques, particularly the van der Pauw configuration are widely adopted because of the effective elimination of the unexpected contact resistances. By contrast, the methodologies for measuring the Seebeck coefficients appear to vary among different research groups. Although their techniques can be in general classified into the differential method and the integral method,^[100] the specific measurement apparatuses reported in most literature are still home-made with customized controls in sample geometries, measurement conditions, etc. By comparing the Seebeck coefficients measured by the set-ups with different geometries, Kemerink and Kemerink reported that an inappropriate geometry setting may induce large systematic errors in the measured values by a factor of >3 .^[101] Later, Pipe and co-workers found that the high environmental humidity may also give rise to an overestimation on the measured Seebeck coefficient than the low humidity conditions.^[102]

In addition to the Seebeck coefficient and electrical conductivity, thermal conductivity is also a critical parameter for the evaluation of ZT . However, it is a difficult task to make a dense block on a millimeter scale for the organic materials.^[103] In a few works, only the through-plane thermal conductivity is measured instead of the in-plane thermal conductivity. As listed in Table 2, the through-plane thermal conductivity of most PEDOT:PSS TE films lie in the range of $0.17\text{--}0.37\text{ W m}^{-1}\text{ K}^{-1}$. Notably, a preferred parallel-to-substrate molecular orientation could be formed during the film formation. The TE parameters including Seebeck coefficient, electrical conductivity and thermal conductivity of the as-obtained PEDOT:PSS films are, therefore, substantially anisotropic rather than isotropic. By applying a time-domain thermoreflectance measurements of the thermal conductivity of the drop-cast PEDOT:PSS films along multiple directions, Liu et al. reported that the in-plane thermal conductivity of a 5% DMSO-added PEDOT:PSS film can reach up to $1\text{ W m}^{-1}\text{ K}^{-1}$.^[104] Wei et al. measured the in-plane thermal conductivity of the 3% EG-added PEDOT:PSS films by using flash analysis methods.^[103] The in-plane thermal conductivity was measured to be $0.84\text{ W m}^{-1}\text{ K}^{-1}$, and the anisotropy factor of the in-plane to through-plane thermal conductivity ($\kappa_{\parallel}/\kappa_{\perp}$) was found to be more than 5. Overall, reliable TE measurement techniques and universal

characterization standards are still highly needed for precisely determining the TE properties of the organic TE thin films.

2.2. PEDOT:PSS-Based TE Composites

Over the past two decades, composites of conducting polymers and inorganic TE materials have become a hotspot in the research field of TE materials.^[105–114] The large interfacial areas and the strong interfacial interactions between the nano-fillers and polymer matrix have been found to yield decoupled TE parameters, which therefore can give rise to significantly enhanced PF and even new functions. Several review articles that cover the recent progress made on the TE composites are highly recommended.^[23,107,115–118] Compared with the individual component, the TE composites commonly possess advantages including filtering out the low-energy charge carriers by interfacial scattering, suppressing the phonon transport by phonon-boundary scattering, and possible ordering the polymer chains by filler templating. It has been highlighted that regardless of the thermal conductivity of the inorganic fillers, the thermal conductivity of the TE composites at low loading can maintain a low thermal conductivity comparable to the polymer matrix due to the phonon scattering effect.^[119] Since PEDOT:PSS has superior solution-processability and is highly compatible with various inorganic materials, the PEDOT:PSS-based TE composites particularly have been the focus of extensive investigation.^[120]

2.2.1. PEDOT:PSS TE Composites with Inorganic Fillers

Inorganic TE materials such as Bi_2Te_3 , lead telluride (PbTe), and tin selenide (SnSe) commonly have PFs on the order of several thousand $\mu\text{W m}^{-1}\text{ K}^{-2}$, which can be nearly one order greater than most organic TE materials. In order to incorporate the intrinsic high TE properties afforded by the inorganic component with the merits, e.g., easy processability, mechanical flexibility, and low thermal conductivity arising from the organic counterparts, inorganic TE materials are widely used as fillers in the organic matrix for the fabrication of TE composites.

Table 3. TE properties of the PEDOT:PSS-based TE composites consisting of inorganic TE fillers.

Fillers	Synthesis method	Loading	$S [\mu\text{V K}^{-1}]$	$\sigma [\text{S cm}^{-1}]$	PF [$\mu\text{W m}^{-1}\text{ K}^{-2}$]	$K [\text{W m}^{-1}\text{ K}^{-1}]$	ZT	Ref.
Bi_2Te_3	Drop casting PEDOT:PSS on Bi_2Te_3 layer	–	145 (p-type)	62	131	≈ 0.5	≈ 0.08	[121]
			–125 (n-type)	50	80			
Bi_2Te_3	Solution mixing	4.1 wt%	16	1295	32.26	0.2	≈ 0.05	[122]
$\text{Bi}_{0.5}\text{Sb}_{1.5}\text{Te}_3$	Solution mixing	30 wt%	130	≈ 5	8.5	0.3	≈ 0.01	[123]
molybdenum disulfide (MoS_2)	Solution mixing	4 wt%	19.5	1250	45.6	0.27		[124]
SnSe	Solution mixing	20 wt%	110	320	380	0.36	0.32	[125]
Te nanowires	In-situ synthesis	–	163	19.3	70.9	0.22–0.30	≈ 0.1	[126]
			180	11	35	0.16	≈ 0.07	[127]
			115	215	284	0.22	0.39	[128]
$\text{Te-Cu}_{1.75}\text{Te}$	In-situ synthesis of inorganic filler in PEDOT:PSS dispersion		220	≈ 2	84	–	–	[129]

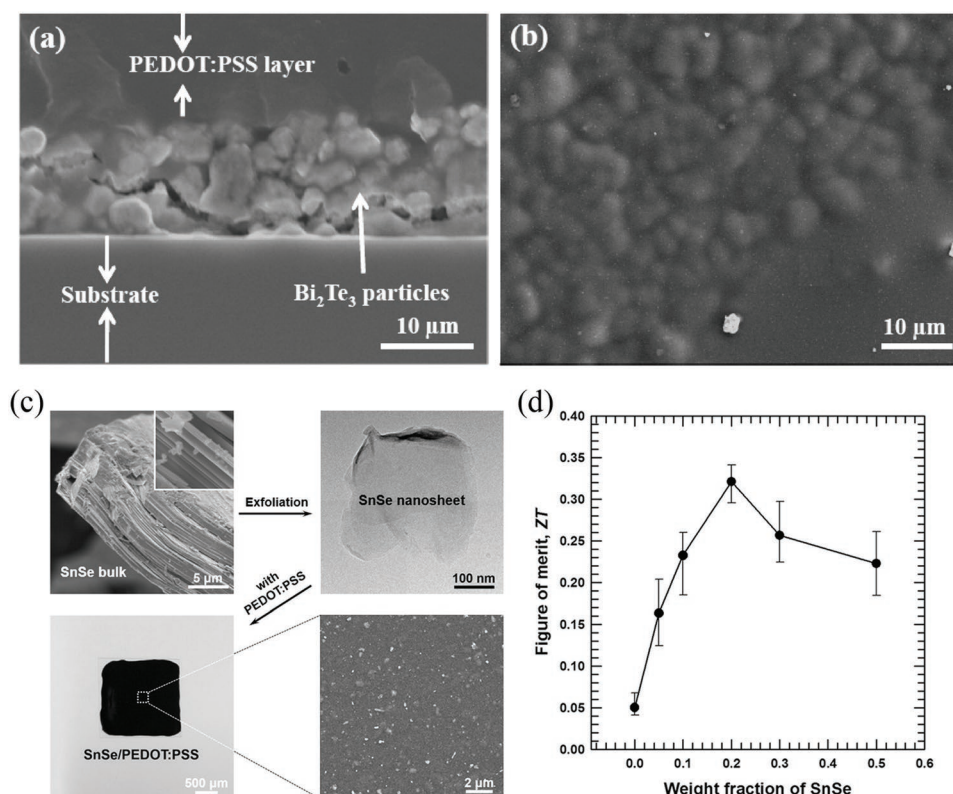


Figure 15. SEM images of the Bi_2Te_3 /PEDOT:PSS composites prepared by a sequential deposition process observed from a) the cross-section view and b) the top view. (a,b) Reproduced with permission.^[121] Copyright 2010, American Chemical Society. c) SEM images and digital image of the SnSe nanosheets and SnSe nanosheet/PEDOT:PSS composite film. d) Power factor of the SnSe nanosheet/PEDOT:PSS composite film as a function of the SnSe nanosheets concentration. (c,d) Reproduced with permission.^[125] Copyright 2016, American Chemical Society.

Table 3 summarizes the TE properties of some PEDOT:PSS-based TE composites with inorganic TE fillers.

Katz and co-workers prepared both p- and n-type TE composites by directly drop casting the 5% DMSO-added PEDOT:PSS dispersion onto the predeposited p- and n-type Bi_2Te_3 films.^[121] Although the PEDOT:PSS dispersion only partially diffused into the Bi_2Te_3 films (as shown in **Figure 15a,b**), the as-prepared p- and n-type Bi_2Te_3 /PEDOT:PSS composites still exhibited significantly enhanced TE properties. The PFs of the p- and n-type Bi_2Te_3 /PEDOT:PSS composites were 131 and 80 $\mu\text{W m}^{-1} \text{K}^{-2}$, respectively, which were 2–3 times greater than that of the neat PEDOT:PSS matrix. Ju and Kim obtained nanostructured SnSe nanosheets from a bulk SnSe by pulverization, Li-intercalation, and subsequent exfoliation.^[125] The SnSe/PEDOT:PSS composites were then fabricated from the mixed solution of SnSe and 5% DMSO-added PEDOT:PSS via drop-casting. For the composites with a SnSe loading of 20 wt%, a maximum PF of 380 $\mu\text{W m}^{-1} \text{K}^{-2}$, and a high ZT of 0.32 were successfully obtained (**Figure 15c,d**).

For the PEDOT:PSS-based TE composites, the incorporation of inorganic fillers can usually give rise to remarkably high Seebeck coefficient. But their electrical conductivities are unsatisfactory. Sometimes the electrical conductivity of the composites could be even lower than any of the individual component.^[121] Recently, various post-treatment methods that can improve the electrical conductivity of PEDOT:PSS were demonstrated for the conductivity enhancement of the PEDOT:PSS-based

TE composites. For instance, Cho and co-workers performed a simple H_2SO_4 treatment on the as-prepared Te/PEDOT:PSS films. After the treatment, the electrical conductivity of the composite films increased from 11 to 215 S cm^{-1} , while the Seebeck coefficient only decreased from 250 to 115 $\mu\text{V K}^{-1}$. Accordingly, the PF was dramatically increased from 69 to 284 $\mu\text{W m}^{-1} \text{K}^{-2}$ ^[128] (**Figure 16**).

2.2.2. PEDOT:PSS TE Composites with Carbonaceous Fillers

Owing to the intrinsic advantages afforded by carbon nanomaterials, the nanostructured carbon allotropes, e.g., carbon nanotubes (CNTs) and graphene which possess high electrical conductivity, large specific area and unique conjugated π - π structure have also been widely employed as fillers in polymer TE composites.^[110,130–135] The TE properties of some PEDOT:PSS-based TE composites with carbonaceous TE fillers are listed in **Table 4**. In the early years, PEDOT:PSS was used as a surfactant for binding the carbon materials into insulating polymer matrix.^[136,137] PEDOT:PSS is able to stabilize the dispersion of CNTs in water and then becomes a part of the polymeric matrix in the composites. Yu and co-workers incorporated CNTs into the an insulating matrix gum Arabic (GA) with the aid of PEDOT:PSS.^[136] When the PEDOT:PSS-coated CNTs form well-percolated networks at 35 wt%, the composites can

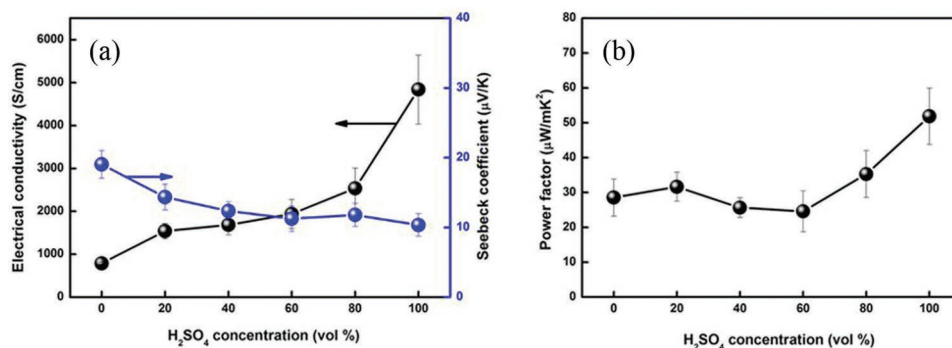


Figure 16. TE properties of the H_2SO_4 -treated Te/PEDOT:PSS composite films. Dependences of a) the electrical conductivity and Seebeck coefficient and b) the power factor of the composite films on the H_2SO_4 concentration. (a,b) Reproduced with permission.^[128] Copyright 2016, Macmillan Publishers Ltd.

have a PF of $25 \mu\text{W m}^{-1} \text{K}^{-2}$. Later, they modified the composites by increasing the loading of PEDOT:PSS and using single-walled carbon nanotubes (SWNTs) instead of multiwalled carbon nanotubes (MWNTs), a Seebeck coefficient of $41 \mu\text{V K}^{-1}$ and an electrical conductivity of 950 S cm^{-1} were obtained for the composite consisting of 60 wt% SWNTs, 30 wt% PEDOT:PSS and 10 wt% polyvinyl acetate.^[137] The PF and ZT were $160 \mu\text{W m}^{-1} \text{K}^{-2}$ and 0.12–0.24, respectively. In spite of the great improvement achieved in PF, the thermal conductivity of the composites still remained to be as low as $0.2\text{--}0.4 \text{ W m}^{-1} \text{K}^{-1}$. The decoupled electrical and thermal properties are attributed to the formation of highly electrically conducting but thermally impeding junctions between the CNTs and PEDOT:PSS.

Post-treatments were also performed on the CNT/PEDOT:PSS composites to enhance their TE properties.^[139–141] Recently, Yu and co-workers reported that the SWNT/PEDOT:PSS films after a treatment with DMSO or FA could achieve PFs of exceeding $400 \mu\text{W m}^{-1} \text{K}^{-2}$ at room temperature.^[139] This PF was over 300 times as that of the untreated ones at the same SWNT loading of 6.7 wt%. However, to maximize the energy filtering effect, appropriate proximity of CNTs should be preserved in the composites. The treatment that can effectively remove PSSH and lead to very close proximity or even direct contact of CNTs may deteriorate the effective interfaces, thus weakening the energy filtering effect.

Chen and co-workers conducted in situ polymerization of PEDOT:PSS films on the template made of reduced graphene oxide (rGO).^[109] Due to the strong $\pi\text{--}\pi$ interaction between rGO and PEDOT, thick PEDOT layers uniformly coated on

both sides of the rGO nanosheets, forming a pie-like rGO/PEDOT:PSS composite structure. As the rGO nanosheets bridge the carrier transport between PEDOT, the carrier mobility was significantly improved. As a result, both the Seebeck coefficient and electrical conductivity were greatly enhanced. The PF was $5.2 \mu\text{W m}^{-1} \text{K}^{-2}$, corresponding to the enhancement by a factor of 13.3 compared with their in situ synthesized PEDOT:PSS. Kim and co-workers reported the in situ synthesis of PEDOT:PSS at the presence of graphene nanoplatelets.^[143] A PF of $45.7 \mu\text{W m}^{-1} \text{K}^{-2}$ was observed for the composite at the graphene loading of 3 wt%, which was twice higher than that of the PEDOT:PSS. Xu and co-workers prepared high-quality graphene nanosheets by liquid exfoliation and fabricated graphene/PEDOT:PSS composites via vacuum filtration.^[141] The composite films were then subjected to a hydrazine treatment. After the treatment, a PF of $53.3 \mu\text{W m}^{-1} \text{K}^{-2}$ was achieved for the composite film with 3 wt % graphene loading, which was five times higher than that of the pristine PEDOT:PSS films. Kim et al. compared the TE performance of the MWNT/PEDOT, graphene/PEDOT:PSS, and the MWNT/graphene/PEDOT:PSS ternary composites.^[144] The ternary composite films showed a PF of $37.08 \mu\text{W m}^{-1} \text{K}^{-2}$, which was superior over either of the two binary counterparts.

2.2.3. PEDOT:PSS-Based TE Composites with Other Polymer Fillers

Significant energy filtering effect has also been demonstrated in the PEDOT:PSS-based TE composites consisting of

Table 4. TE properties of the PEDOT:PSS-based TE composites consisting of carbonaceous fillers.

Fillers	Synthesis method	Loading [wt%]	$S [\mu\text{V K}^{-1}]$	$\sigma [\text{S cm}^{-1}]$	PF [$\mu\text{W m}^{-1} \text{K}^{-2}$]	$K [\text{W m}^{-1} \text{K}^{-1}]$	ZT	Ref
SWNTs	Solution mixing	60	41	950	160	0.2–0.4	–	[137]
SWNTs	Solution mixing	85	≈ 20	≈ 3300	≈ 140	0.45–0.90	≈ 0.03	[138]
SWNTs	Solution mixing then DMSO treatment	6.7	59	≈ 1350	464	–	–	[139]
DWNTs	Solution mixing then EG treatment	20	43.7	780	151	–	–	[140]
rGO	In situ polymerization	16	31.8	50.8	5.2	–	–	[109]
Liquid exfoliated graphene nanosheets	Solution mixing then hydrazine treatment	3	22	1283.5	53.3	0.3	0.05	[141]
Graphene quantum dots	Solution mixing then EG/DMSO	5	14.6	72	1.5	–	–	[142]
Graphene nanoplatelets	In situ polymerization	3	26.8	637	45.7	–	–	[143]

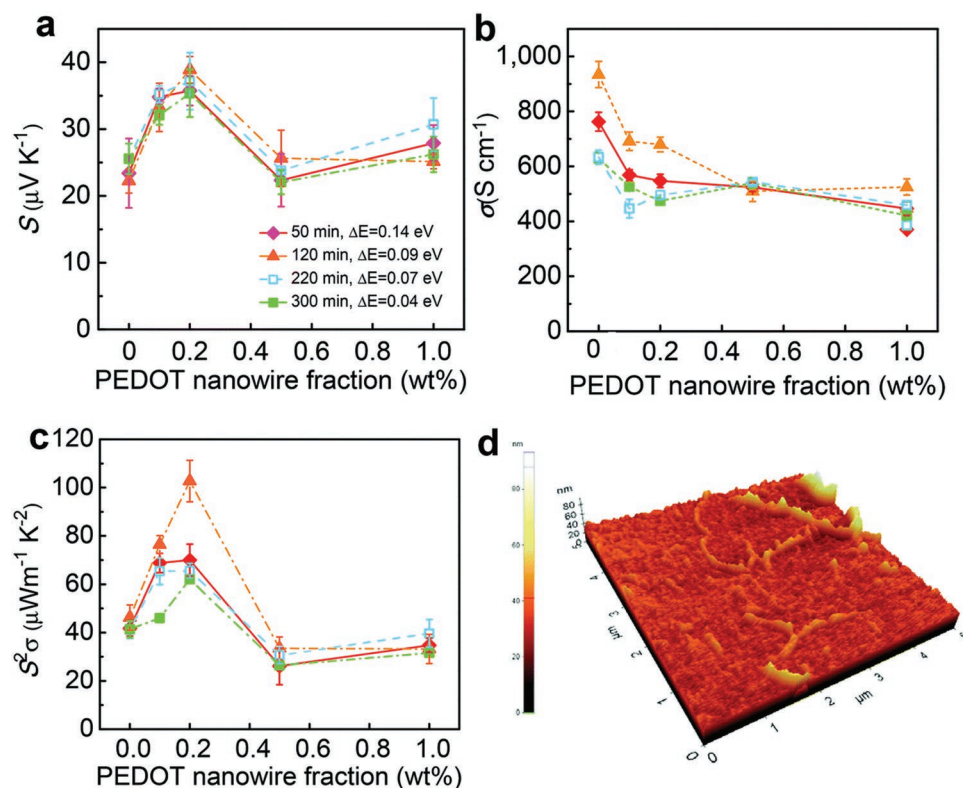


Figure 17. TE properties of the PEDOT nanowire/PEDOT:PSS composite films. a) Seebeck coefficient, b) electrical conductivity, and c) power factor of the composite films as a function of the PEDOT nanowire fraction. d) The AFM image of the 1 wt% PEDOT nanowire/PEDOT:PSS composite film. All panels reproduced with permission.^[146] Copyright 2016, Royal Society of Chemistry.

nanostructured polymers.^[145–148] Wang and co-workers demonstrated the PEDOT nanowire/PEDOT:PSS composites by a solution-mixing process,^[146] for which the PEDOT nanowires was obtained from an aqueous solution via a self-assembly approach.^[147] As shown in **Figure 17**, the Seebeck coefficient of the PEDOT nanowire/PEDOT:PSS composites exhibited strong dependence on the nanowire loading. As the PEDOT nanowire fraction increased from 0 to 0.2 wt%, the Seebeck coefficient were enhanced from 22 to 38.9 $\mu\text{V K}^{-1}$. Since the electrical conductivity only slightly decreased, the PF was improved to 102.7 $\mu\text{W m}^{-1} \text{K}^{-2}$, corresponding to an enhancement by nine-fold. The great TE property enhancement of the all-PEDOT composite is attributed to the synergetic effects of interfacial energy filtering and change of carrier concentration induced by the addition of PEDOT nanowires in the PEDOT:PSS matrix.

A PEDOT:PSS/PANi TE composite with novel multilayered structures was demonstrated by Jo and co-workers.^[149] The PEDOT:PSS with 5% DMSO addition and PANi-camphorsulfonic acid (CSA) solutions were alternatively deposited on a glass substrate via an LBL technique (**Figure 18a**). With the increase in the deposition repetition cycle increased, the electrical conductivity of the multilayer composite films increased. At the deposition repetition cycle of 4, the electrical conductivity and PF reached values of 1550 S cm^{-1} and 56 $\mu\text{W m}^{-1} \text{K}^{-2}$, which were 1.3 and 2 times higher than that of the neat PEDOT:PSS, respectively (**Figure 18b,c**). The transmittance electron microscopy, electron energy loss spectroscopy, and Raman analyses revealed that the multilayer structure of

PEDOT:PSS and PANi induced stretching of PEDOT and PANi chains in the composite, which in turn facilitated the effective hole diffusion between the PANi to PEDOT:PSS layers.

3. PEDOT:PSS-Based TE Devices

3.1. TE Generators (TEGs)

The availability of high-quality PEDOT:PSS aqueous dispersion enables the facile fabrication of PEDOT:PSS-based devices via various solution-processing techniques. In the small laboratory scale, spin-coating, and drop-casting are the most common methods for the preparation of PEDOT:PSS films used for material characterization. But for constructing a practical TE module, scalable and high-efficient processing techniques that can be applied in the industrial scale must be seriously accounted. Up to now, various printing techniques have been reported for the fabrication and/or assembly of the PEDOT:PSS-based TE modules. The schematic diagrams of some representative printing techniques and typical TEG module structure are illustrated in **Figure 19**. The as-fabricated TE modules can be utilized as wearable TEGs and TE sensors in practice, suggesting great application in the fields of wearable and self-powered electronics.

Wei et al. made an all-organic TE module with the 5% EG-added PEDOT:PSS solution on a paper substrate via screen-printing, as shown in **Figure 20**.^[150] The individual PEDOT:PSS

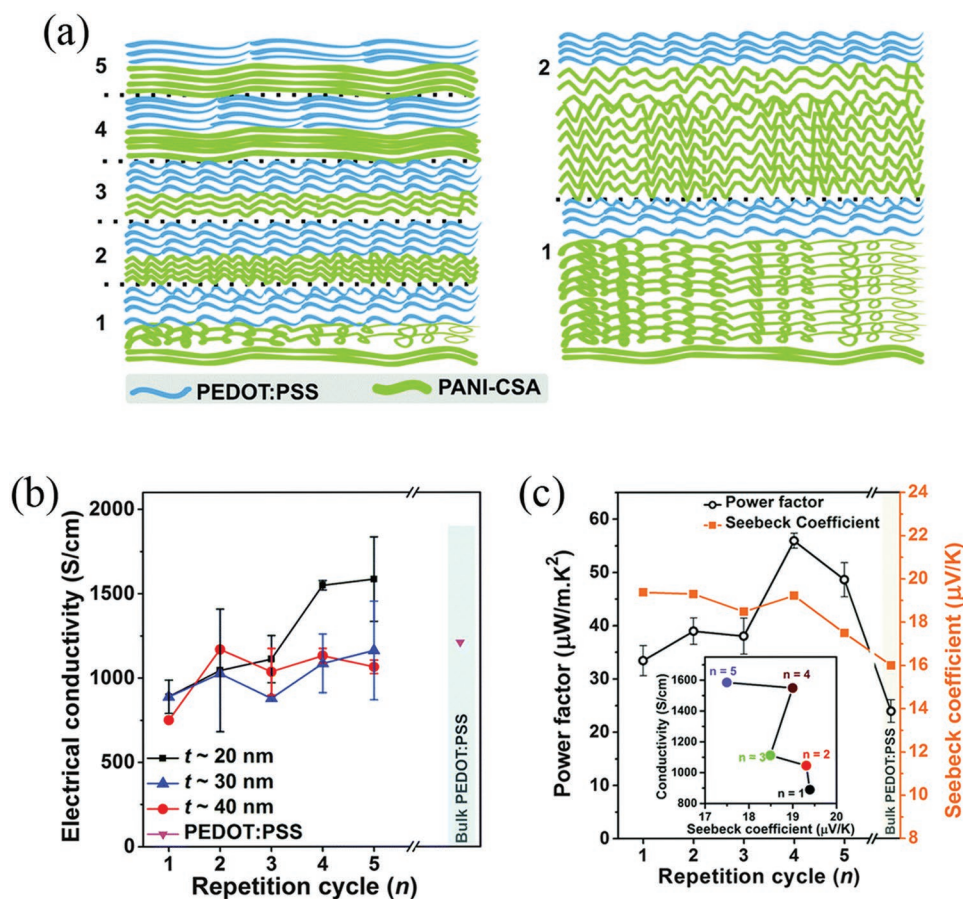


Figure 18. a) Schematic diagram of the chain stretching mechanism for the multilayered PEDOT:PSS/PANI composite film. b) Electrical conductivity and c) power factor of the composite film as a function of the repetition cycle. All panels reproduced with permission.^[149] Copyright 2016, Royal Society of Chemistry.

TE leg had PF of a $25 \mu\text{W m}^{-1} \text{K}^{-2}$ with a Seebeck coefficient of $18 \mu\text{V K}^{-1}$ and an electrical conductivity of 770 S cm^{-1} . Silver paste was then applied to connect the 11 individual TE legs on the same paper substrate. A prototype TEG was built by connecting a few pieces of such TE paper either in series or in parallel with metal wires. The whole stack was finally sandwiched between two copper plates for effective heat conduction. The TEG containing 70 pieces of TE paper (each 5 pieces connected as one group in parallel and 14 groups in total connected in series) delivered a maximum output power of $4 \mu\text{W}$ at $\Delta T = 50 \text{ K}$.

Wang and co-workers proposed an R2R printing technique for continuously printing the solution-processable polymeric TEG on a flexible substrate.^[151] As shown in Figure 19c, the R2R printing system was composed of three sequential rollers, which were employed for subsequently printing the p-type, n-type, and connection strips onto the substrate. For each single pair of the rollers, the ink was kept in an epoxy-coated reservoir, and the ink transfer cylinder was halfway immersed in the reservoir, facilitating the ink transfer from the reservoir to the printing rollers. During the printing process, the printing rollers contacted with the substrate under a small pressure, so that the ink was transferred onto the substrate. As the substrate was propelled through the system at a fixed rate, the TEG could be successfully produced with the desired patterns. By

such a simple process, a TEG consisting of PEDOT:PSS as the p-type TE legs and nitrogen-doped graphene as the n-type TE legs was successfully fabricated, as shown in Figure 21. For an R2R-printed TEG film with dimensions of $8 \text{ cm} \times 10 \text{ cm}$, the maximum output power reached $\approx 0.24 \text{ mW m}^{-2}$ at $\Delta T = 10 \text{ K}$.

As a versatile approach, all the TE pastes for TE module printing should be readily applicable to be coated on the conventional fabrics and textiles. Wu and Hu developed a TE composite of waterborne polyurethane (PU), MWNTs, and PEDOT:PSS via solution mixing. Before drying, the composite mixture was paste-like. The textile yarns, e.g., polyester filament and cotton yarns were dipped into the mixture for several cycles to achieve uniform coating. Due to the fabric nature, the as-coated TE yarns retained flexibility as good as the uncoated counterparts (Figure 22). Their power factor, however, was only $1.4 \mu\text{W m}^{-1} \text{K}^{-2}$ at room temperature, which still needed to be much improved to meet the requirements of a practical wearable TEG.^[155]

3.2. TE Sensors

TE materials with high Seebeck coefficients can be versatility applied as temperature sensors based on the Seebeck effect.^[156] Carella and co-workers made an all organic p–n

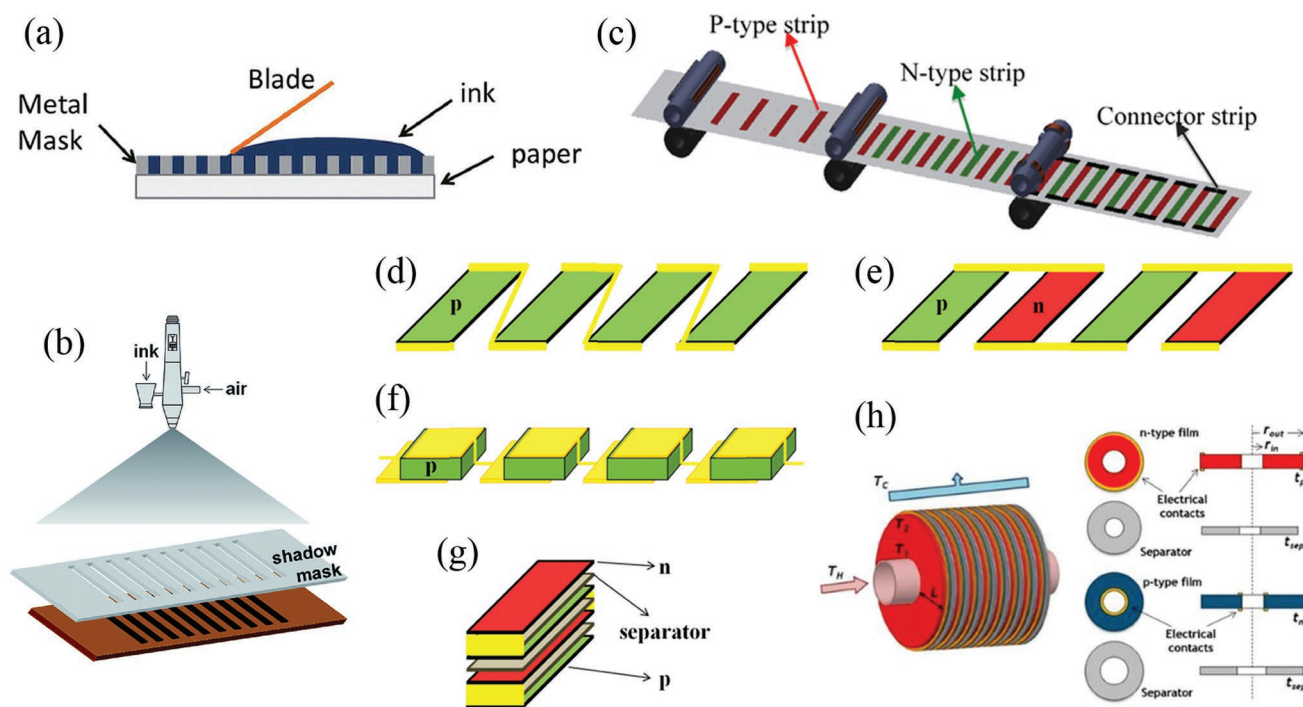


Figure 19. Schematic diagrams of some representative printing techniques a) screen-printing. Reproduced with permission.^[150] Copyright 2014, Royal Society of Chemistry. b) Spray printing. Reproduced with permission.^[134] Copyright 2015, Royal Society of Chemistry. c) Roll-to-roll printing. Reproduced with permission.^[151] Copyright 2016, Elsevier. Some representative structural design of the TEG module (planar configurations consisting of d) p-type only and e) p- and n-type TE films, f) vertical configuration with p-type films, g) multilayered configuration with p- and n-type TE films,^[152,153] and h) radial configuration with p- and n-type films.^[154] Reproduced with permission.^[118] Copyright 2017, Elsevier.

TE sensor on the polyethylene naphthalate substrate using PEDOT:PSS film and nickel ethylenetetrathiolate coordination polymer poly[$K_x(Ni-ett)$] as p- and n-legs, respectively. In order to enhance the Seebeck coefficient of PEDOT:PSS, the PEDOT:PSS films were treated with EG and TDAE in sequence. At the optimal condition, the p- and n-legs

showed Seebeck coefficients of 161 and $-160 \mu V K^{-1}$, respectively. As a result, the sensitivity of this sensor was even higher than that of the reference made of metals.^[62]

Temperature sensors are important for many areas. For instance, multifunctional sensors and electronic skins (E-skins) request not only for precise temperature sensing but also

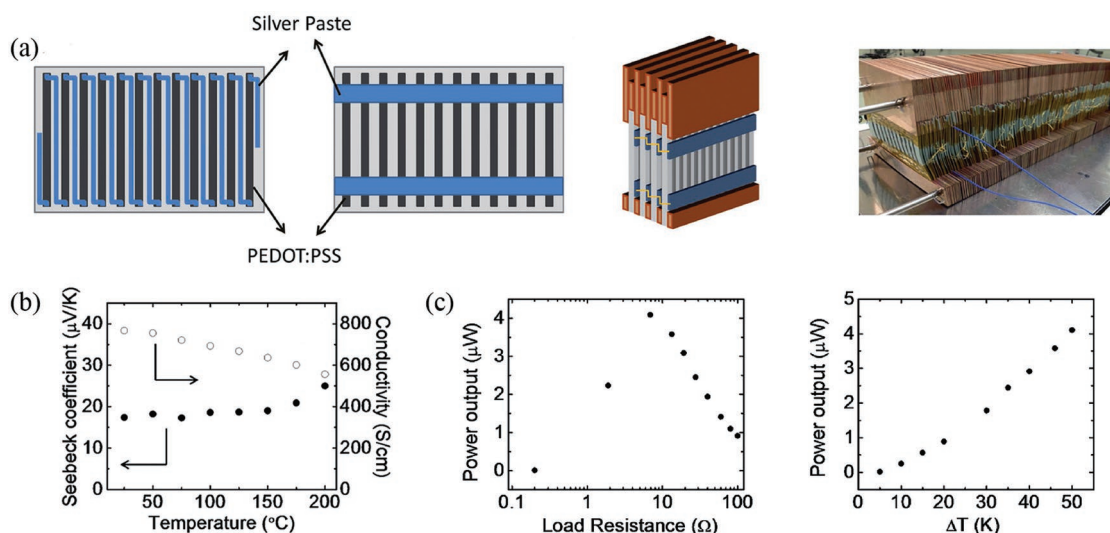


Figure 20. a) Schematic diagram and digital image of the screen-printed PEDOT:PSS TE module on paper substrate. b) TE property of the 5% EG-added PEDOT:PSS TE leg. c) Output power of the TE module containing 70 pieces of TE paper. All panels reproduced with permission.^[150] Copyright 2014, Royal Society of Chemistry.

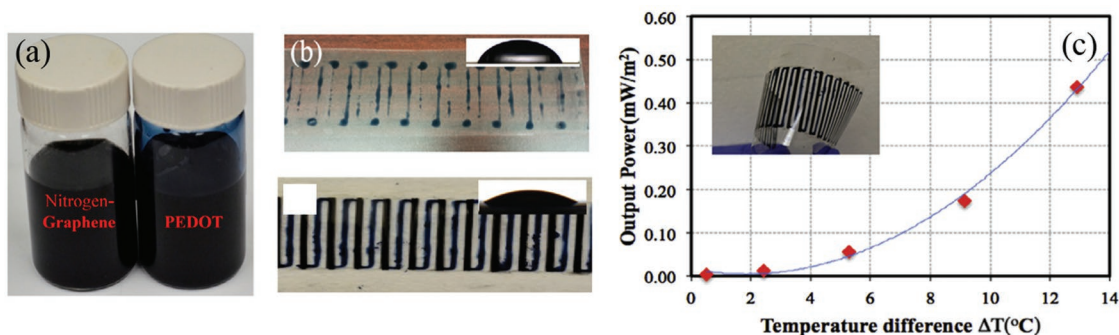


Figure 21. a) The PEDOT:PSS and nitrogen-doped graphene inks used for R2R printing. b) Digital image of the TE module fabricated by the R2R printing technique. c) The output power of the R2R-printed TEG at different ΔT . All panels reproduced with permission.^[151] Copyright 2016, Elsevier.

demand for additional functions, e.g., simultaneous strain sensing and self-powering. Zhu and co-workers developed a flexible temperature-pressure dual functional sensor by depositing PEDOT:PSS on a microstructured PU frame^[157] (Figure 23). The PEDOT:PSS possessed a Seebeck coefficient of $\approx 30 \mu\text{V K}^{-1}$ and facilitated the voltage (V) shift upon the current-voltage (I - V) curve. When the device experienced a temperature change on its upper surface, a voltage (V_{thermal}) could be generated due to the Seebeck effect, $V_{\text{thermal}} = \text{Seebeck coefficient} \times \Delta T$ with ΔT being the difference between the temperature and room temperature. Owing to the porous structure of PU frame, the pressure-induced strain led to corresponding resistance change upon pressure stimuli. As the temperature- and pressure-sensing were governed by the TE property of PEDOT and the supporting frames of PU, respectively, the sensor transduced temperature and pressure stimuli as independent signals. Such a temperature-pressure dual functional sensor featured a pressure sensitivity and temperature-sensing accuracy of $>20 \text{ kPa}^{-1}$ and $<0.1 \text{ K}$, respectively. Given that a thermovoltage V_{thermal} was provided during the temperature sensing, the dual-functional sensor could be self-powered by this V_{thermal} as the input source. Accordingly, the pressure could be detected by the $\Delta I/I_0$ ($I_0 = V_{\text{thermal}}/R_0$) signal without any other power supply. The application of such a flexible temperature-pressure dual-parameter sensor as electronic skins was further demonstrated, as shown in Figure 23b. The sensor was fabricated into a $500 \mu\text{m}$ thick semitransparent fabric glove with 12×12 pixels via a simple stamp printing or

inkjet printing technique. Upon the contact of either a warm hand or an ice cube, the sensor array showed both an excellent spatial resolution and dual-parameter-sensing capability.

4. Summary and Outlook

Over the past decades, significant progress has been achieved on TE polymers and TE composites based on PEDOT:PSS. PFs of several hundreds to nearly one thousand $\mu\text{W m}^{-1} \text{K}^{-2}$ have been reported for the PEDOT:PSS after a secondary doping, chemical treatment, sequential treatment or in the heterostructure configuration with the presence of an ionic conductor. Despite these exciting achievements, the development of organic TE materials is still in the initial stage. The PFs and ZT values reported for the PEDOT:PSS-based TE materials are still far below the requirements for practical applications. In order to further improve their TE performance, the features including high carrier mobility, well ordered packing of polymer chains with longer conjugated length, desirable morphology, and appropriate doping levels should be achieved at the same time. Creative approaches for material synthesis, morphology control, structural design, and effective post-treatment routines are therefore highly required. For the PEDOT:PSS-based composites, the selection of filler materials, composite fabrication techniques, rational interfacial engineering, and theoretical models for explaining the great TE property enhancement still need to be optimized. More possible applications based on the PEDOT:PSS TE polymers and

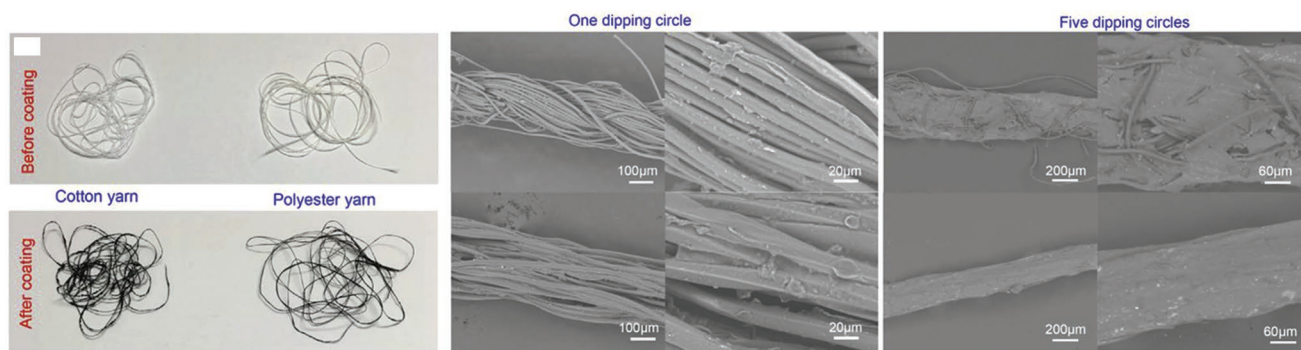


Figure 22. Digital images and SEM images of the of the cotton and polyester yarns after PEDOT:PSS/PU coating. Reproduced with permission.^[155] Copyright 2016, Elsevier.

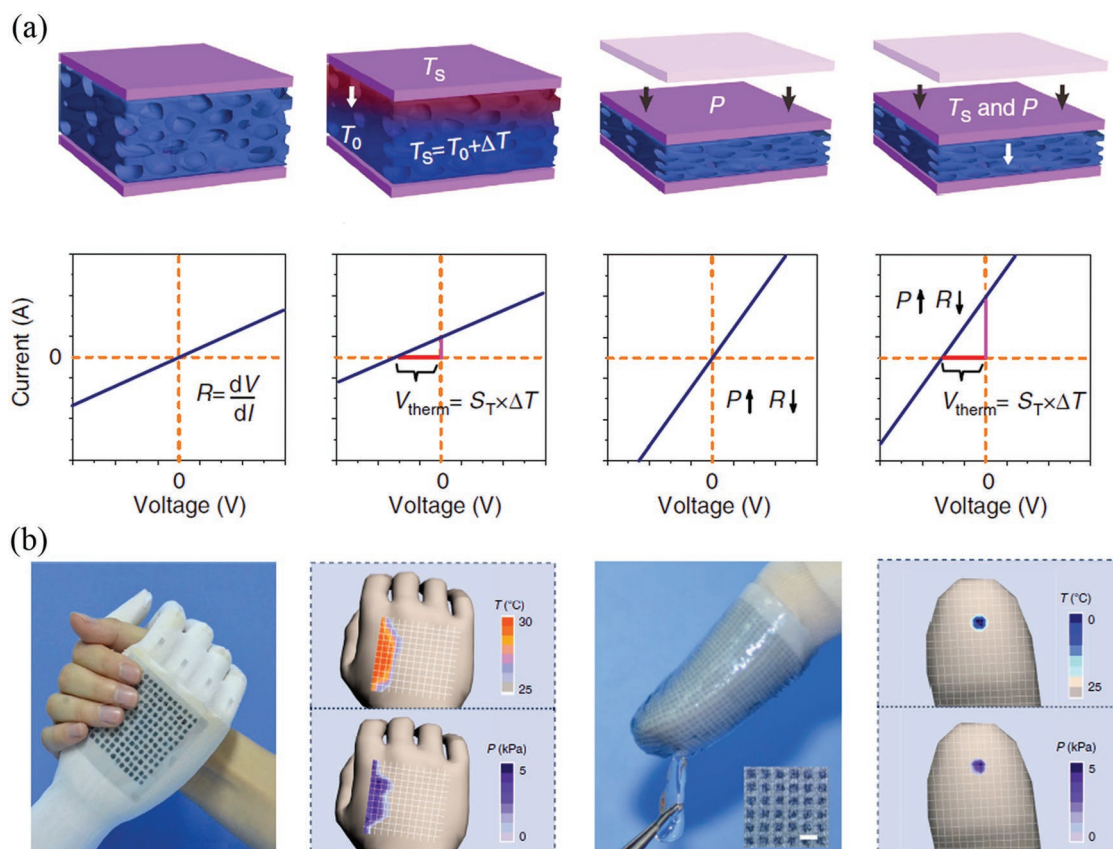


Figure 23. a) Schematic illustration of temperature-pressure sensing mechanism of the microstructure-frame-supported organic thermoelectric (MFSOTE) device and its corresponding signals upon stimuli: initial state, upon ΔT , pressure, and a coupled temperature and pressure stimuli. b) Photographs and sensing mappings of the self-powered MFSOTE matrixes under a coupled temperature and pressure stimuli. Reproduced with permission.^[157] Copyright 2015, Macmillan Publishers Ltd.

composites are also greatly anticipated. The integration of TE technology with other energy conversion or storage techniques may open new opportunities for the wide applications of the PEDOT:PSS-based TE materials in the future.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

electrical conductivity, PEDOT:PSS, power factor, Seebeck coefficient, thermoelectric polymers

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